

ACHIEVE!



Geography

Sustainability in the Asia Pacific

Jennifer Leigh-Jones



Made for the
Australian Curriculum



Achieve! Geography

Get ready for the Australian Curriculum with editable work sheets that have been written to assist teachers to embed the cross-curriculum themes of Indigenous histories and cultures, Australia's place in the Asian region, and sustainability.

Features

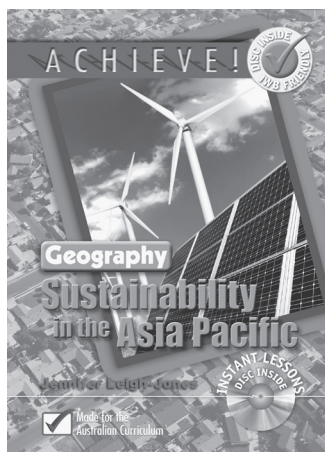
- ✓ Activities designed to reinforce and enhance literacy skills
- ✓ Emphasis on understanding essential geography concepts and content
- ✓ Comprehensive Teacher's Notes that provide clear lesson objectives, background information, starter and extension activities
- ✓ CD-ROM containing all pages from the book making them ideal for use on IWBs
- ✓ Word files on the CD-ROM that can be edited to suit the specific needs of students



Software requirements

Windows – MS Word 97 or higher, Adobe Acrobat Reader

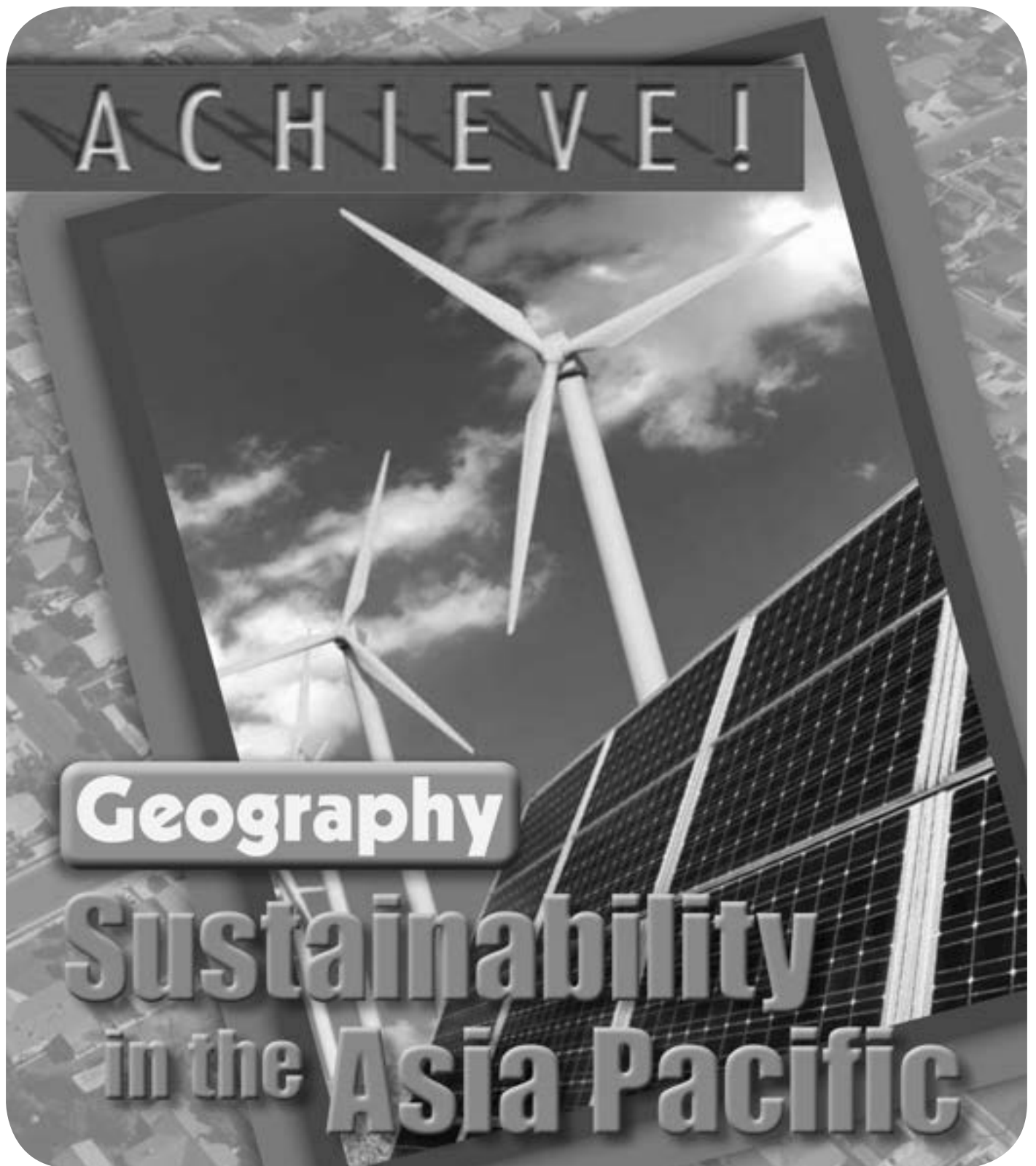
Mac – MS Word 98 or higher, Adobe Acrobat Reader



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The *Achieve!* series of Geography titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students who struggle to read and comprehend regular classroom materials because of poor literacy skills. Many of the pages will also be suitable as supplementary material for regular classes.

Sustainability in the Asia Pacific will assist teachers to develop lessons regarding the cross-curriculum priorities – *Sustainability*, and *Asia and Australia's engagement with Asia* – from the Australian Curriculum. Students develop their geographic understanding and inquiry skills as they learn about alternative sources of energy production including solar, hot rock, and wave power. The work sheets also focus on how communities in the Australasian region respond to these new production forms as well as the impact they have on the environment and peoples' everyday lives.

The CD-ROM contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other work sheets. These documents can be easily edited to suit the individual needs of students.

This book is divided into ten units, each of which generally contains one or more Resource sheets and several Activity sheets. This will allow teachers to use the material in a variety of ways. For example, the unit could be taught as a lesson with students in groups of varying sizes. Alternatively, a single Resource sheet and related Activity sheets can be used as support material, for individual homework, assessment or revision.

The teacher's notes provide valuable guidance about using each unit, and contain the following elements.

- **Objectives** – These are the key skills and knowledge learnt through using the work sheets.
- **Prior knowledge** – This refers to the skills and knowledge required for students to complete the tasks. Some activities are more challenging than others. However, as a rule, students should have a reading comprehension age of six to nine years to be able to attempt the activities.
- **Background** – This section provides additional guidance to help teachers present a successful lesson. It may include definitions or further background notes about the subject matter.
- **Starter activity** – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.
- **Resource and Activity work sheets** – The Resource sheets do not always contain activities and are used either to stimulate discussion or as a part of the unit's activities. The Activity work sheets may be used sequentially or as stand-alone tasks, depending on the needs of the student.
- **Extension ideas** – Additional tasks are suggested to recap on the main skills or knowledge taught in the lesson, as well as ways to extend, assess or include technology in the lesson.

Objectives

- To understand the concept of sustainability
- To recognise the need for sustainability
- To identify renewable and non-renewable resources
- To learn how sustainable energy can assist with climate change and the health of the environment

Prior knowledge

- Students should have some understanding of resources and the fact that they can be used up, especially as the population grows. They should also know that the use of fossil fuels has an impact on climate change, air quality and health, and that it is possible to generate power from renewable energy sources.

Background

There has been much discussion regarding the need to use renewable energy as our understanding of the effects of burning fossil fuels increases, and our fossil fuel supplies decrease. However, all forms of renewable energy have advantages and disadvantages. Wind power, solar power, tidal power, geothermal energy and biomass are regarded as green energy sources. Hydro-electric power and nuclear power are considered sustainable energy sources by some people.

Starter activity

Compile a class list of all the ways students use power. Then introduce the idea that power is also used to produce and distribute everything they buy. Add to the list the things the students buy regularly.

Resource and activity sheets

Provide students with the Activity sheet, 'Sustainability'. Read the introduction and speech bubbles with the students. Encourage them to suggest further examples of sustainability, then discuss what sustainability is. Students will work in pairs to write their own definitions for sustainability.

Read the Resource sheet, 'Why do we need sustainable energy?', with the students. For each fossil fuel, point out the number of years it is expected to last at current rates of consumption. Encourage students to suggest reasons why countries still use fossil fuels. Introduce the idea that it is cheaper in terms of money for countries to continue using fossil fuels than to invest in the research, technology and infrastructure need to use renewable energy sources.

Ask students to complete the Activity sheet, 'Global supplies of fossil fuels'. They will need a copy of the Resource sheet, 'Why do we need sustainable energy?', an atlas or access to an Internet map search, and a blank world map. A map is provided on the CD-ROM in the Artwork > Unit 1 folder. Some students may need assistance with developing an appropriate key. When they have completed the mapping activity, students answer questions about the regions that have large and small fossil fuel reserves.

Read the Activity sheet, 'Renewable and non-renewable resources' with the students. Ensure students understand the meaning of 'resource' and that resources can be renewable or non-renewable. Introduce the idea that while some resources are renewable, they can also be scarce, like bananas after a cyclone in Queensland. Also, some resources require sound environmental management to be used sustainably; for example, fish supplies can be unsustainable due to human activities such as ocean pollution and overfishing. Students could use the Internet to check their answers to the last question.

Have students complete the Activity sheet, 'Reducing energy use'. Some students might prefer to work in pairs. You could compile a class list of their ideas for reducing energy use.

Extension idea

- Take a tour of the school, and have the students note ideas for reducing energy use. If possible, choose several ideas and develop a plan to implement them.

Sustainability

The population of the earth has grown very rapidly to 7 billion people. This has placed a huge demand on the earth’s resources, which are mostly used to generate power. As countries such as China and India become more developed, there is a greater demand for resources – especially power. How can this growing demand be met?

What does sustainability mean to you?



As a builder, I want to build houses that can be used for a long time, and are designed to keep warm in winter and cool in summer.



I’m a farmer and I want to keep planting and harvesting food crops, and sell them for a fair price.



I work as an irrigator, providing extra water for farmers’ crops. I want fresh water to be available now and in the future.



I’m a wind farm manager and I want more of our power to come from the wind.



I want to know my clothes are made in a factory that has safe working conditions and obeys pollution laws.



I live near a river – I don’t want the water to be polluted by chemicals, or to be used up by irrigation.

-  1 Discuss with a partner your ideas about **sustainability** and what it means. Then write your own definition for sustainability.

Why do we need sustainable energy?

Oil, coal and natural gas are fossil fuels. They formed over millions of years from the remains of plants and animals.

In 2010, 80.9% of global energy use was supplied by fossil fuels, and in Australia, 94.8% was supplied by fossil fuels.

Burning fossil fuels for energy creates pollution and releases carbon dioxide into the air, which contributes to climate change.

Fossil fuels are non-renewable resources, and they are starting to run out.

Top 10 countries: Proven oil reserves (thousand million barrels)	
Venezuela	296.5
Saudi Arabia	265.4
Canada	175.2
Iran	151.2
Iraq	143.1
Kuwait	101.5
United Arab Emirates	97.8
Russian Federation	88.2
Libya	47.1
Nigeria	37.2
Enough for 51 years at 2011 rate of consumption.	

Top 10 countries: Proven coal reserves (million tonnes)	
United States	237 295
Russian Federation	157 010
China	114 500
Australia	76 400
India	60 600
Germany	40 699
Ukraine	33 873
Kazakhstan	33 600
South Africa	30 156
Colombia	6 746
Enough for 112 years at 2011 rate of consumption.	

Top 10 countries: Proven natural gas reserves (thousand million barrels)	
Russian Federation	44 600
Iran	33 100
Qatar	25 000
Turkmenistan	24 300
United States	8 500
Saudi Arabia	8 200
United Arab Emirates	6 100
Venezuela	5 500
Nigeria	5 100
Algeria	4 500
Enough for 65 years at 2011 rate of consumption.	

Source: BP Statistical review of world energy, 2012

Global supplies of fossil fuels

To complete this activity, you need the Resource sheet, ‘Why do we need sustainable energy?’, an atlas or Internet map search, and a blank world map.

-  1 Locate the top 10 countries from each table on the Resource sheet, ‘Why do we need sustainable energy?’ and label them on the map.
- 2 Work out a key to show which fuel – oil, coal, natural gas – each country has.
Note that some countries have more than one type of fuel.
The key you use will need to allow for this.

Use your map to answer the following questions.

- 3 Which region has the most oil? _____
- 4 Which region has the least oil? _____
- 5 How might this affect the cost of oil in the future for many countries?

- 6 Which region has the most coal? _____
- 7 Which region has the least coal? _____
- 8 Which region has the most natural gas? _____
- 9 Which region has the least natural gas? _____
- 10 How might this affect the cost of energy in the future for many countries?

Renewable and non-renewable resources

 **1** Unjumble and write correctly the words in brackets to complete these definitions.
The first letter is underlined.

- a) A **resource** is something that we can use for our _____ (ftniebe).
- b) A **renewable** resource can be replaced by _____ (unlarta) ecological cycles or by good environmental _____ (nnmgemetaa).
- c) A **non-renewable** resource cannot be _____ (plreedca) – it is _____ (noge) forever once it has been _____ (sude).

 **2** Are the resources listed in the box renewable or non-renewable? Write them in the correct part of the table.

● bananas	● eggs	● gold	● sugar	● uranium
● beef	● fish	● iron	● sunlight	● wheat
● coal	● fresh water	● ocean waves	● timber	● wind
● copper	● gas	● oil	● tomatoes	● wool

Renewable	Non-renewable

3 Which of the resources listed above can we use for energy production?

Reducing energy use

 **1** Use the word bank to complete the information below.

Using renewable _____ – that is, energy that does not give out carbon _____ – can help to reduce the amount of _____ gases in the atmosphere. This will help stop further warming of the earth. The use of renewable energy such as wind power, _____ power, geothermal energy and wave power should be encouraged.

Using _____ energy will help the environment – and _____ our energy bills! That is why energy efficiency is important. We can _____ our houses to reduce the amount of energy used for heating and cooling. We can use double glazing and shades to keep _____ in, and carefully choose where to plant trees to help keep heat out. We can save energy by wearing a jumper in _____ rather than turning up the heater.

It is important that we _____ about our lifestyles and the changes we can make to reduce energy use. We can ride a bike or walk or even use _____ transport to get to school or work, instead of driving cars. Using hybrid cars and electric cars helps to reduce the amounts of _____ fuels we use.

When we shop, we can buy products that are made _____ and don't have to be transported by plane, ship or _____ over long distances. We can grow our own vegetables – then they don't have to be transported at all. It is also important to plant _____, which take in _____ dioxide and give out _____.

2 How can you, your family and your school reduce the amount of energy you use?
Write at least one idea in each box.

Me	My family	My school

Word bank

carbon

dioxide

energy

fossil

greenhouse

heat

insulate

less

oxygen

public

reduce

solar

think

trees

truck

winter

Objectives

- To learn about wave power
- To consider the advantages and disadvantages of using wave power, now and in the future
- To consider the effects of technological development on power generation

Prior knowledge

Students should have an understanding of the need to use clean, renewable energy, now and in the future.

Background

The idea of using wave power has been around for at least 200 years. The technology is now available and has been proven to be a viable source of power, although there is concern about the longevity of the equipment. As with all new technology, the cost of research and development and the initial infrastructure costs are significant factors. However, as fossil-fuel-generated power becomes more expensive and society focuses on reducing carbon emissions, this method of power generation will become more affordable.

Australia is well placed to utilise wave power with its vast coastline and huge oceans washing the shores. Wave power sites include Geraldton in the west, along the south coast, in Tasmania and on the northern coast of New South Wales.

In the Asia Pacific, Japan is seeking safer alternatives to nuclear power and the government is investigating ways to use their 4.47 million square kilometres of territorial waters for power generation.

Starter activity

Have students share their experiences of the ocean. Guide the discussion to bring out the ideas that waves are continuous, and they are powerful. Raise the idea of using the power of the ocean to generate electricity. Encourage students to suggest how they think this could work.

Resource and Activity sheets

Read the Resource sheet, 'What is wave power?', with the students. You could provide images of coastal features such as the 12 Apostles on the Victorian south coast to extend the idea of waves shaping coastlines over time. Tell the students that wave energy has also been responsible for many shipwrecks, both near coasts and out at sea. Look carefully at the diagram showing how wave power is harnessed. Point out that as well as harnessing energy, this technology can supply another scarce resource – fresh water.

Have students complete the cloze exercise on the Activity sheet, 'Generating power'. Encourage them to consider the role of technology in providing renewable energy.

Provide students with the Resource sheet, 'Opinions about wave power'. Take the opportunity to discuss the difference between an opinion and a fact. Students could decide what their own opinions are about wave power.

On the Activity sheet, 'Advantages and disadvantages', students summarise what they have learned about wave power, and categorise their knowledge into advantages and disadvantages. Students may find it useful to refer to the Resource sheets, 'What is wave power?' and 'Opinions about wave power'. Then students will write a persuasive letter to an electricity supply company.

Extension idea

- Students could design a survey to find out what people know about wave power. They could also find out whether people think it is a good idea or not, and the reasons for their opinions.

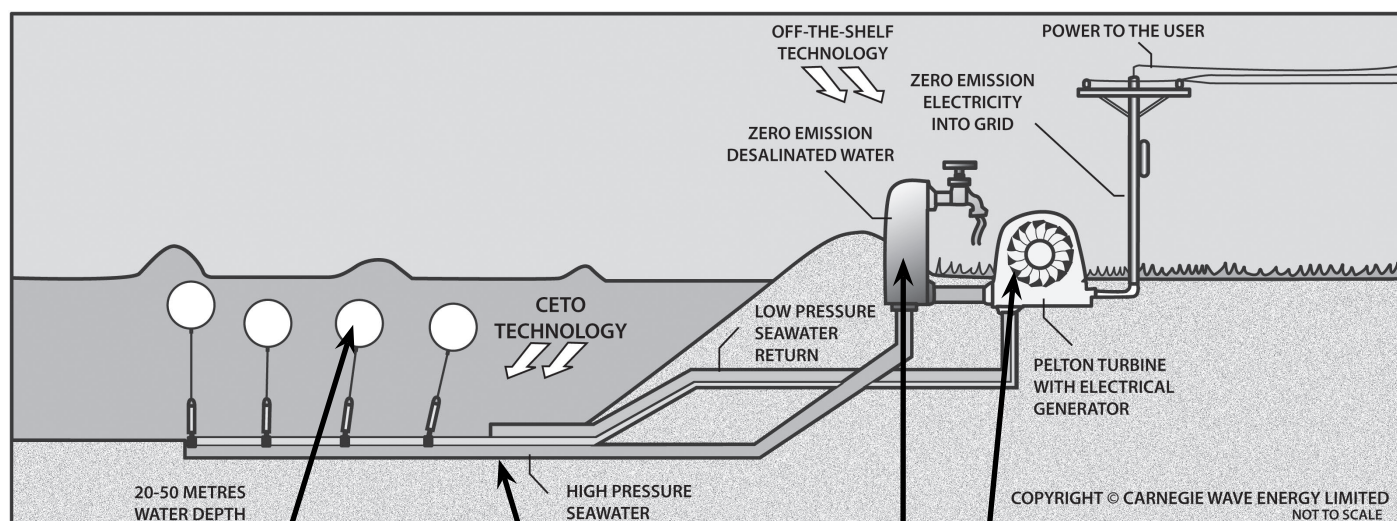
What is wave power?

Every day a huge amount of **energy** is created by waves pounding the Australian coasts. This energy is largely responsible for shaping our coastlines. As renewable energy becomes more important, experiments are being conducted to see if electricity can be produced **economically** from this source. Wave power is **continuous** and does not produce **greenhouse gas emissions**.

Wave power captures energy from the surface waves, unlike tidal power which uses the ebb and flow of the tides. The first wave power farm was built in Portugal in 2006. It can produce electricity continuously.

The CETO wave power farm is based in Perth, Western Australia. The diagram below shows how it works.

The CETO wave power converter



Reproduced with permission

These buoys are powered by the waves to drive pumps that are anchored to the sea floor.

The pumps pressurise the water, which then passes through a pipeline to the shore.

The high-pressure seawater then drives turbines, which generate electricity without producing emissions.

As well as generating electricity, the wave power converter can remove the salt from seawater so it can be used for drinking or irrigation like any other fresh water.

Wave power in Australia

 1 Use the word bank to complete the information below.

The use of _____ power to generate _____ is a relatively new technology in Australia. At the moment, it costs more to generate electricity using waves than it does to generate using _____. But it does work – in 2010, an experimental wave power plant in Port Kembla in NSW successfully supplied power to the _____ for three months.

However, fossil fuel power is becoming more _____ and people are considering other factors such as _____ carbon emissions. As _____ advances, wave power is likely to become more affordable. Construction began on a _____ wave energy project near Perth, Western Australia, in January 2010. This system uses buoys to capture wave energy. The buoys drive _____ that are anchored to the sea bed. High pressure water travels from the pumps by _____ to power generator turbines on the shore. When it is completed, it will be the first wave power system in Australia to be _____ permanently to the power distribution grid.

In August 2010, a similar system was announced for Eden in NSW. A commercial plant could be operating there within three to five years.

Portland, Phillip Island and Warrnambool in Victoria; Geraldton and Albany in Western Australia; and Port MacDonnell in South Australia are all being considered as suitable _____ for wave power generation.

Using wave power has several key advantages:

- _____ are continuous and reliable.
- Because so many Australians live near the _____, suitable wave energy locations are close to the people who will actually _____ the power.
- Due to advances in technology, wave power can be _____ by waves 1 to 2 metres in height, which increases the number of possible locations.

Word bank

coal
coast
commercial
connected
electricity
expensive
generated
grid
pipes
pumps
locations
reducing
technology
use
wave
Waves

Opinions about wave power



Wave energy power will mean jobs – I would like to live near the coast.



I like the fact that wave power is renewable and does not produce any waste or release any emissions into the atmosphere.



It is important to me that this kind of power can be supplied 24 hours a day.

For wave power to be work, the power plant must be where waves are strong, and the equipment must be able to stand up to rough weather.



As a fisherman, I worry that the wave power buoys will make it hard to move in and out of port.



My house overlooks the coast. I don't want my view ruined by power lines.



I'm worried that the equipment might affect the fish and other wildlife in the area.

 1 Read the Resource sheets, 'Using wave power' and 'Opinions about wave energy'. Work with a partner to discuss the advantages and disadvantages of using wave power. Record your ideas in the table below.

2 Write a short letter to your local electricity supply company to persuade them to support the use of wave power.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The paper has a slightly irregular, hand-drawn border. There are approximately 20 lines across the page. The top-left corner is folded over.

Objectives

- To learn about geothermal energy
- To understand the difference between natural geothermal energy and engineered geothermal energy
- To appreciate the advantages and difficulties associated with geothermal energy

Prior knowledge

Students should understand that the earth has layers and the earth's core is molten. They should be aware that erupting volcanoes spew out hot lava.

Background

In the past, geothermal energy was only harnessed in volcanic regions where naturally occurring hot water and steam came to the surface. For centuries Japan and New Zealand have used this resource for cooking. More recently it has been used to generate electricity and pipe steam to heat houses, and for tourism attractions.

Engineered thermal energy is a renewable resource capable of producing base load (24 hours per day) energy. It does not release carbon dioxide into the atmosphere. No waste is produced during the process. The environmental impact is minimal – drill holes, pipelines and power plants.

Potentially, geothermal energy could produce enough energy to supply Australia for a very long time. However, the necessary infrastructure of powerlines is not currently in place and would be very expensive. Small earthquakes can occur as a result of the drilling – one plant in Switzerland had to be shut down after earthquakes caused damage to buildings. Obtaining the information needed to map suitable sites is also very expensive.

Starter activity

Before class, conduct an Internet image search or YouTube search for 'geothermal mud pools'. Select images and /or movies to show the class. Display them, and ask students to explain what that think is happening. Some students may have visited the mud pools in Rotorua, New Zealand. Tell the students that this is an example of heat and steam coming from within the earth's surface.

Resource and Activity sheets

Read the Resource sheet, 'What is geothermal power?', with the students. Ensure the students are familiar with the meaning of each word shown in bold. Discuss the difference between natural geothermal energy and engineered geothermal energy. To extend the activity, you could watch the Internet video about Paralana geothermal plant (Internet search terms: 'Paralana geothermal plant').

Provide students with the Activity sheet, 'Hot rocks'. Review the concept of the layers of the earth. You could use the analogy of an egg: the earth's crust is like the shell of the egg, the mantle is like the egg white and the core is like the egg yolk. Allow time for students to practise and revise their verbal explanations for how geothermal power is generated. This particular diagram shows a binary cycle geothermal power plant – further information about how it works is readily available on the Internet.

Read the Resource sheet, 'Advantages and disadvantages', with the students. Discuss the items in the lists and ensure students understand why each one is either an advantage or a disadvantage. You could work with the class to compare these advantages and disadvantages to those of another type of energy source, such as coal, wave or solar.

Have students complete the Activity sheet, 'Hot rock energy production'. They may need to refer to the Resource sheets, 'What is geothermal power?' and 'Advantages and disadvantages'. After they have completed the activity, students could work in small groups to discuss their ideas for the second section.

Extension idea

- Have the students cut out or draw the diagram from the Activity sheet, 'Hot rocks'. Challenge them to label the diagram to explain what is happening and how it works. Students may like to work in pairs. The diagram shows a binary cycle geothermal power plant. Further information about the use and function of these power plants is readily available on the Internet.

What is geothermal power?

Geothermal power is a form of renewable energy that comes from heat in the earth's crust. *Geo* means 'earth' and *thermal* means 'heat'. For every 100 metres down into the earth you go, the temperature increases by about 3°C.

Natural geothermal energy

- **Natural** geothermal energy is produced when water seeps into the earth's crust, where it is heated by hot **volcanic rocks** and then rises to the surface as **steam**.
- In North America, **hot springs** have been used for cooking for thousands of years.
- In countries such as Indonesia, the Philippines, China, Japan and New Zealand this **resource** has been used for heating, power generation and cooking.
- Five geothermal energy plants are planned for Fergusson Island, which is part of the Papua New Guinea **archipelago**.
- In the Philippines, geothermal power provides 27% of the country's total electricity production.

Engineered geothermal energy

- Australia has a vast amount of **hot granite** within the earth directly beneath it. This could be used to produce power by injecting water into breaks in the rocks, which is then returned to the surface as steam. This is called **engineered geothermal energy** because it involves using technology to produce energy from heat deep inside the earth.
- South Australia, in particular, has great potential for this form of renewable energy. A small geothermal plant capable of generating 520 mega-watts of energy has been built at Paralana next to the Beverley Uranium Mine. A plant at the Cooper Basin also shows a promising future in hot rock geothermal energy. There are thought to be enough hot rocks in South Australia to provide power for all of Australia for a very long time. However, the set-up costs for geothermal plants, such as the costs of drilling, are high.

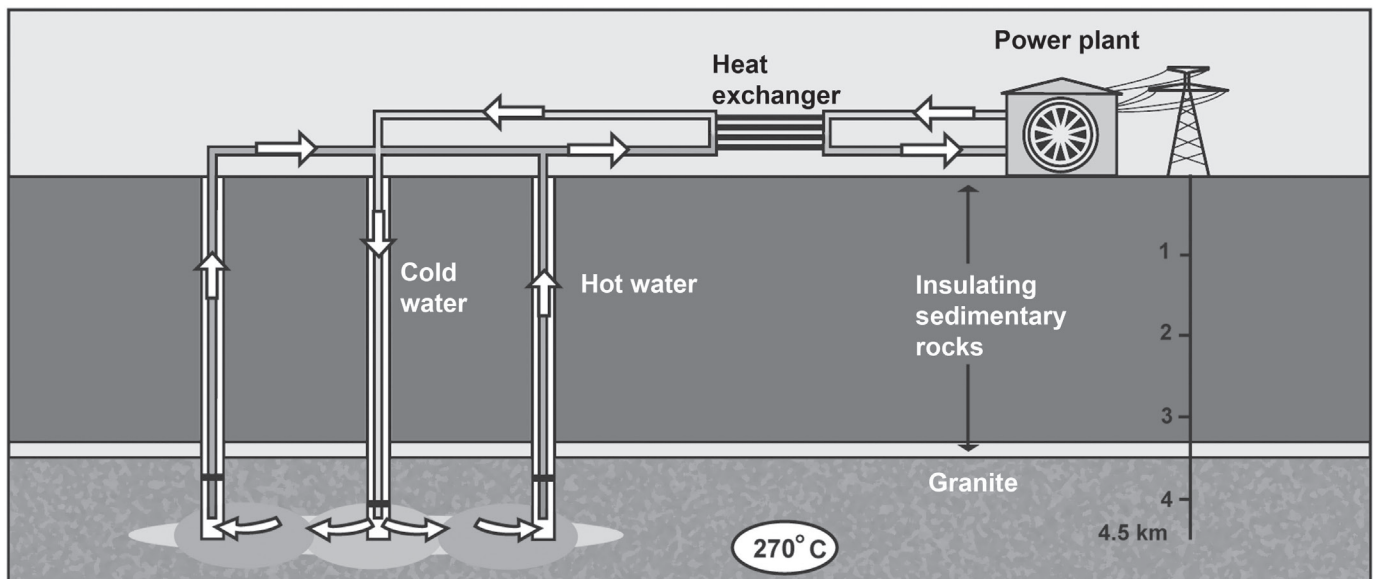
Hot rocks

Millions of years ago, the granite rocks in the **Cooper Basin** in South Australia were on the surface of the earth. Over time, they were covered by a layer of **sedimentary** rocks that is now four kilometres thick. This layer insulates the granite, which absorbs heat from deep within the earth. The granite four kilometres below the surface has a temperature of more than 250°C.

 **1** Unjumble and write correctly the words in brackets to complete these rock definitions.

- **Granite rocks** are a type of igneous rock. They were formed by _____ (mmaag) trapped _____ (woebl) the earth's surface, which slowly _____ (doelco).
- **Sedimentary rocks** are also called secondary rocks. They were _____ (fmerdo) on the earth's _____ (ascfuer) from _____ (seepci) of other rocks, or organic material such as _____ (snobe) and _____ (lehssl).

How to generate geothermal power



- 2** Look carefully at the diagram above which shows how geothermal power is generated. Using the Resource Sheet, 'What is geothermal power?', to help you, explain to a partner how it works.

Advantages and disadvantages

Advantages

- There is no emission of carbon dioxide.
- No waste is produced.
- Energy can be produced all the time.
- Levels of energy produced can be increased within minutes to meet higher demand for power.
- It can be used for heating at the site where the power is generated.
- Production of geothermal energy creates jobs.
- It is a sustainable form of energy.

Disadvantages

- It can be expensive to set up if it is a long way from major centres.
- Temperatures increase with depth, but so does the cost of drilling.
- Little earthquakes can be caused by drilling through the rocks.
- It is expensive to map the locations of hot rocks.

Hot rock energy production

-  **1** Use the Resource sheets, 'Geothermal power' and 'Advantages and disadvantages', to help you decide whether the statements below are true or false.

Hot rock energy production	True	False
There are no carbon emissions.		
It creates jobs.		
It is cheap to set up.		
Temperatures stay the same when you drill into the earth's crust.		
It is expensive to map the location of hot rocks.		
It produces a lot of waste.		
Power cannot be produced all the time.		
The hot water can be used for spas.		

- 2** Write three paragraphs to explain what you think of the idea of using hot rocks for generating power in the future.

I think...

I also believe... _____

Finally... _____

Objectives

- To learn about wind power and its advantages and disadvantages
- To learn how and where wind power can be generated
- To understand different opinions about wind farms

Prior knowledge

Students should be aware that wind is an integral part of weather and climate, and be able to give examples of the use of wind in daily life.

Background

As the demand for energy increases with a growing global population, fossil fuel resources are dwindling. Increased energy consumption means carbon dioxide and other greenhouse gas emissions are contributing to climate change. In an effort to reduce greenhouse gas emissions, renewable energy sources such as wind are being explored. Governments are setting targets to encourage the use of clean, renewable energy such as wind power, although it cannot generate a base load (24-hour) supply.

Starter activity

Introduce the concept of the wind. Discuss with the students what it is, and how people harness the power of the wind. As well as its potential to generate energy, encourage students to consider its role in sailing, in sports such as hang gliding and kite surfing, and its effect of different kinds of aircraft. Point out that people cannot control the wind, and discuss its potential to cause damage; for example, storms and cyclones and spreading bushfires.

Activity sheets

Provide students with the Activity sheet, 'Wind power'. Students will need an atlas, a blank map of Australia and Internet access to complete this activity. A map is provided on the CD-ROM in the Artwork > Unit 4 folder. Before students begin the activity, make sure they understand the meaning of each term in bold. When they have finished, discuss the distribution of the wind farms and any patterns in the kinds of places they are located. To extend the activity, students could research global use of wind power. Countries such as New Zealand and China use wind-generated power.

On the Activity sheet, 'Wind power and sustainability', students will extend their knowledge about wind power, and the importance of using renewable energy.

The Activity sheet, 'Carbon dioxide emissions', provides a graph of carbon dioxide emissions per person in Australia. Students are guided through the process of analysing the graph, then explaining what this means in terms of future energy use.

On the Activity sheet, 'How wind turbines work', students will gain an understanding of how wind power is generated by completing the description of how each part works.

The Activity sheet, 'Current use of wind power', provides information about wind power, and presents a pie graph showing the capacity for wind power electricity generation in Australia, for each state and territory. Have the students consider the proportions in terms of physical size and population of each state, and other natural resources it may be more efficient to harness. Students will answer questions about the graph, and list advantages and disadvantages of wind power.

Read the Activity sheet, 'Opinions about wind farms', with the class, and ensure they understand the meaning of each term in bold. Then students write their own opinions about wind farms. Encourage them to provide reasons for their opinions.

Extension idea

- Ask students to design a survey to find out what people know about wind farms, and their opinions on whether wind farms are a good idea or not.

Wind power

Wind is a **renewable energy resource** – this means that it does not produce **carbon dioxide** or any chemicals that affect public health or the environment. The use of renewable energy is an important factor in reducing the impact of **climate change** and in making sure we can continue to produce energy in the future.

By world standards, Australia has excellent wind resources. The southern coastline lies in the **roaring forties** – between the latitudes of forty and fifty degrees. In this area, strong winds blow much of the time, because there is little land in the Southern Hemisphere to stop them.

 **1** You need a map of Australia to complete this activity.

- Shade and label the map to show the roaring forties region.
- Label the map to show the wind farms listed below. There is a capacity in megawatts (MW) listed for each farm. Develop a key to identify the capacities.

New South Wales:

Blayney (9.9 MW)
Capital wind farm
(140.7 MW)
Crookwell (4.8 MW)
Cullerin Range (30 MW)
Hampton (1.32 MW)

Victoria:

Challicum Hills
(52.5 MW)
Codrington (18.2 MW)
Portland (132 MW)
Toora (21 MW)
Waubra (192 MW)
Wonthaggi (12 MW)

South Australia:

Canunda (46 MW)
Cathedral Rocks (66 MW)
Clements Gap (56.7 MW)
Hallett (298.2 MW)
Lake Bonney (278.5 MW)
Mount Millar (Yabmana)
(70 MW)
Snowtown (98.7 MW)
Starfish Hill (34.5 MW)
Waterloo (111 MW)
Wattle Point (90.8 MW)

Queensland:

Windy Hill (12 MW)

Western Australia:

Nine Mile Beach (3.6 MW)
Albany (21.6 MW)
Emu Downs (79.2 MW)
Hopetoun (1.2 MW)
Kalbarri (1.6 MW)
Ten Mile Lagoon (2 MW)
Walkaway (89.1 MW)

Tasmania:

Huxley Hill (1.7 MW)
Woolnorth (Bluff Point)
(139.75 MW)

Note: More wind farms are in the planning stages or under construction

Wind power and sustainable energy

Wind power uses the power of the wind to drive a **turbine**. This produces electricity that can be connected to the **power distribution grid**.

Wind farms are located in **reliably** windy areas, usually on farmland or along the coast. The wind does not blow all the time so other energy sources must be available.

-  1 Decide whether these statements are true or false.
Then correct any false statements to make them true.

	True	False
The power of the wind drives a turbine to create electricity.	☐	☐
Reliable wind means wind that blows regularly.	☐	☐
Coastal areas are not windy.	☐	☐

- 2 Use the word bank to complete the information about the importance of finding ways to produce sustainable energy in Australia.

Word bank

planting emissions wind cars renewable carbon
dioxide greenhouse equivalent

Sustainable electricity from _____ power helps Australia cut its _____ gas _____ by almost 3 256 000 tonnes of _____ per year. This is _____ of removing 752 000 _____ from the roads, or _____ 4.86 million trees.

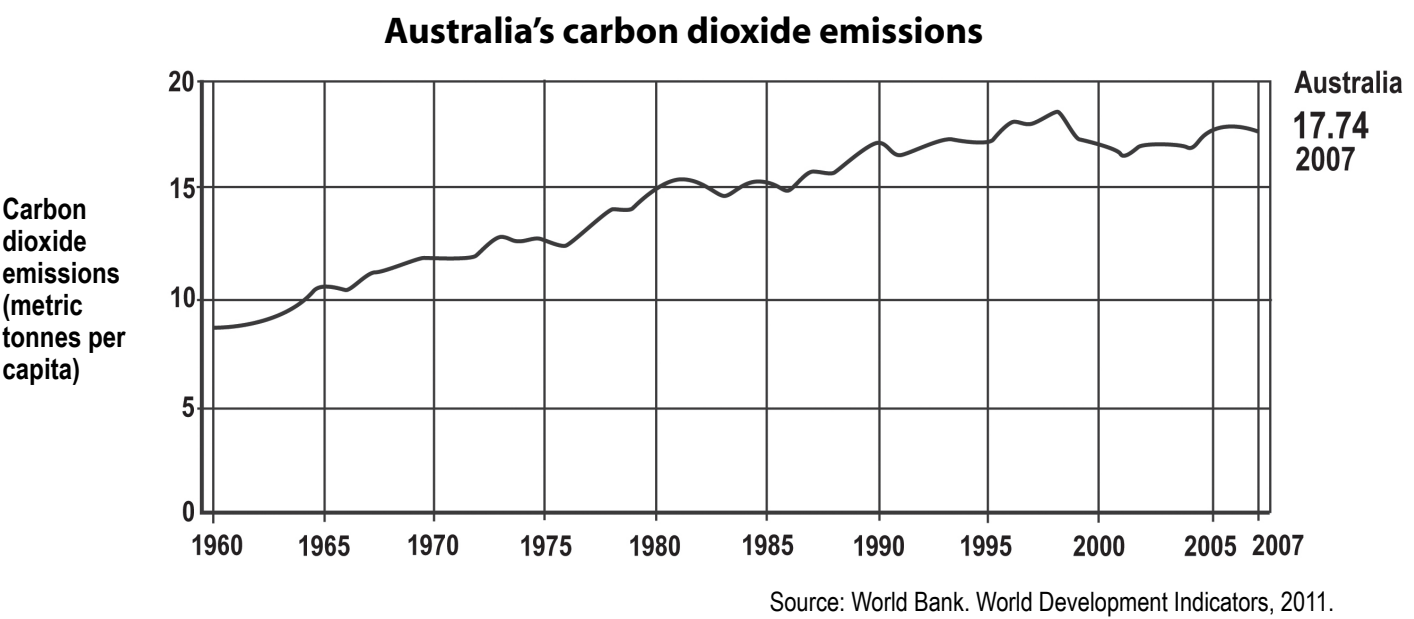
The Australian government is aiming to produce 20% of our energy from _____ sources by the year 2020.

- 3 In the letter string below:
- circle the Asia Pacific country where wind power is generated
 - underline two Australian towns that have wind farms.

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Carbon dioxide emissions

Look at the line graph showing carbon dioxide (CO₂) emissions per person (per capita) in Australia and answer the questions below.



- 1 Suggest why the measurements are given **per capita** (that is, per person) and not as total amounts.
-
- 2 What is the **trend** of emissions between 1960 and 2007?
-
- 3 Which year had the highest emissions? _____
- 4 How much carbon dioxide was emitted per person in 2001? _____
- 5 What change occurred between 2005 and 2007? _____
-
- 6 Write dot points explaining what the graph tells you about the need for renewable, clean energy sources like wind power.
- _____
 - _____
 - _____

How wind turbines work

-  **1** Read the information below and look at the diagram. Complete the words by putting in the correct vowels.

W _ n d t _ r b _ n _ s:

Change the energy of the
wind into electricity

Bl _ d _ s:

Catch the wind

G _ _ rs:

Are moved to transmit
energy

G _ n _ r _ t _ r:

Captures the energy and
passes it to the power grid

T _ w _ r:

Holds up the turbine

Gr _ d c _ n n _ ction:

At the base of the tower

F _ _ nd _ t _ _ ns:

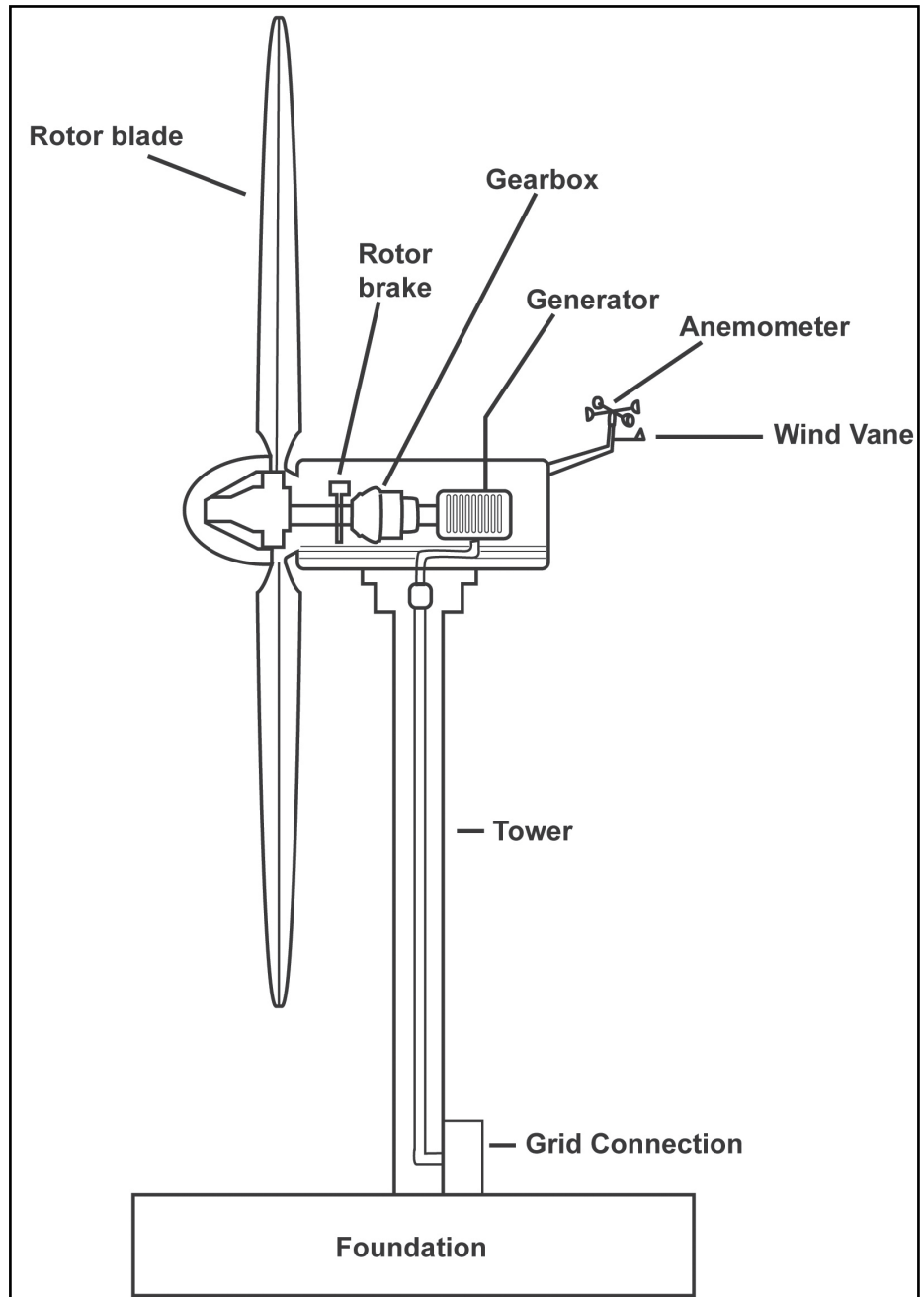
To support the tower

An _ m _ m _ t _ r:

Measures the wind speed

W _ n d v _ n _:

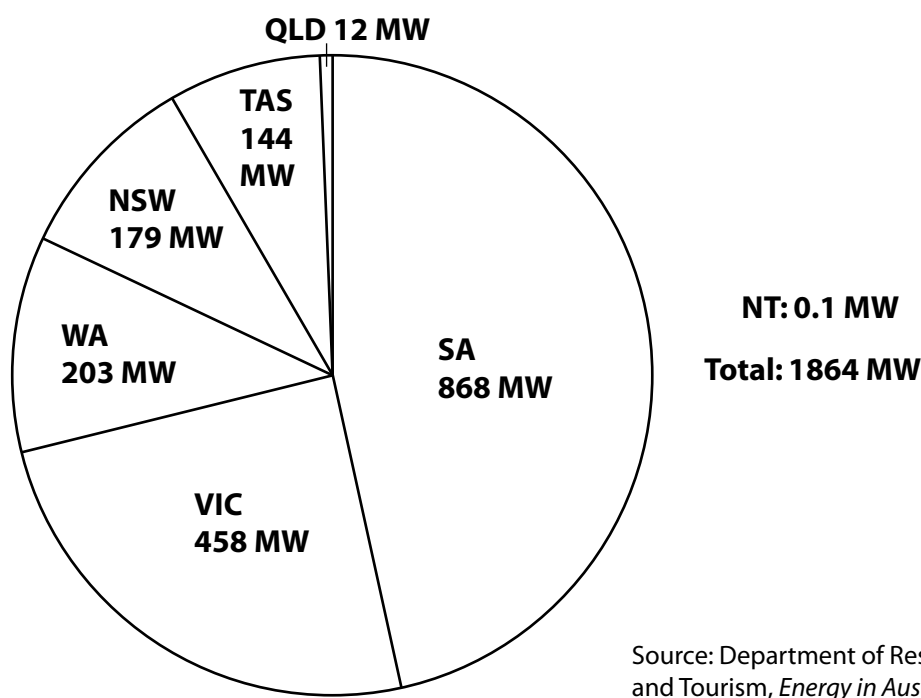
Shows the direction the
wind is blowing from



Current use of wind power

Manufacturing wind turbines and setting up wind farms creates **employment** during construction, as well as ongoing **maintenance** jobs, especially in rural areas. Farmers receive **income** for putting turbines on their land. **Stable prices** are another benefit, as electricity generated by wind power is not subject to the same price changes as oil. Oil prices can increase due to cold winters in Europe and the USA or wars in major oil-supply regions.

Power generated by wind farms in Australia, by state, 2010



Use the pie chart to answer the following questions.

- 1 Which state has the most wind energy capacity? _____
- 2 Which state has the least wind energy capacity? _____
- 3 Suggest why some states use more wind power than other states.

- 4 List three benefits of wind power.

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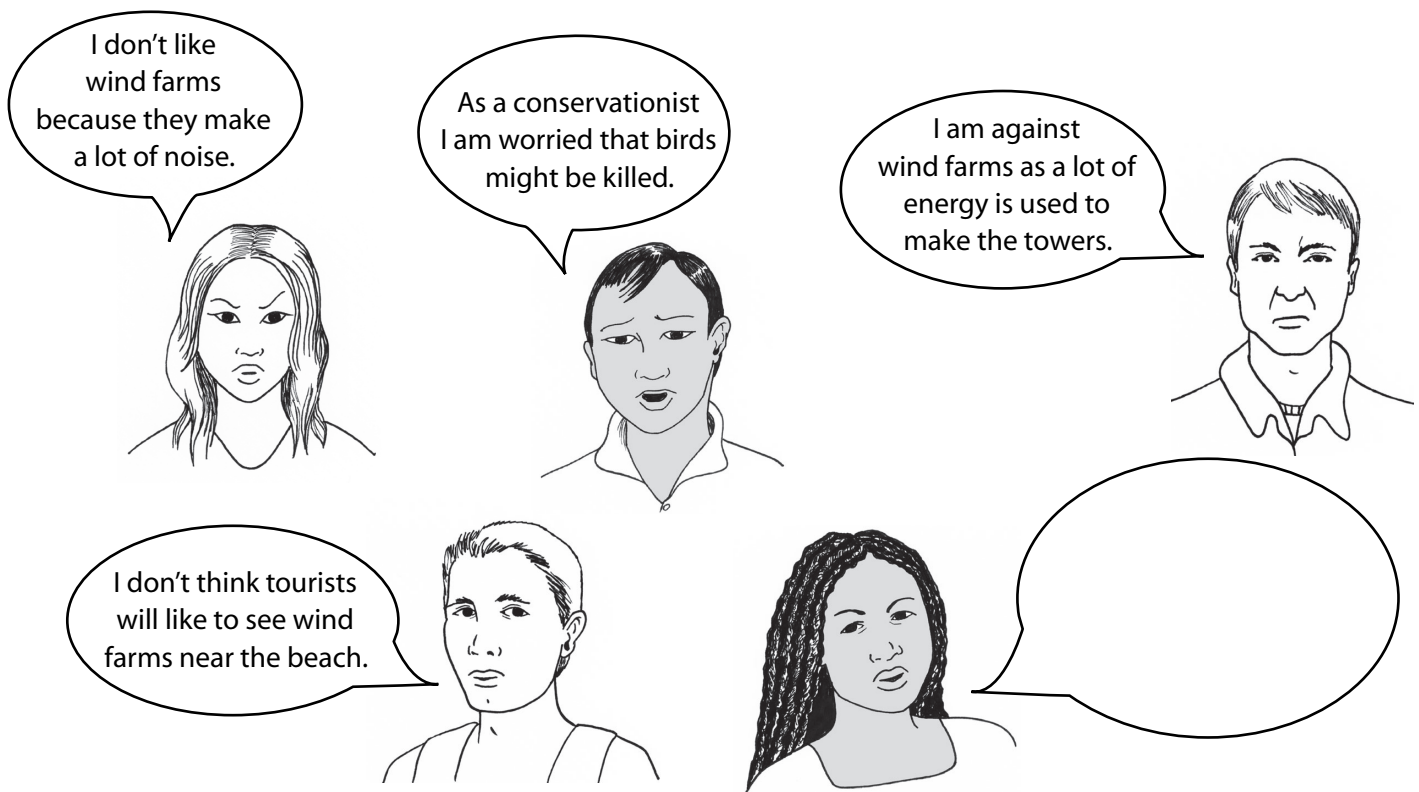
Opinions about wind farms

Not everyone agrees that wind farms are a good idea. Some people do not like the look of wind farms as they are often located in places that are considered areas of beauty, such as near the coast or on hills. But are they worse than large power lines?

The wind farm **Waubra**, 35 km north-west of **Ballarat** in Victoria, consists of 128 turbines. There have been claims of illness caused by the low-level noise of the blades turning. Some residents have reported suffering headaches, high blood pressure, rapid heartbeats and nausea. One person moved to Ballarat to get away from the noise. His son operates the farm but will not live there.

Some people are concerned about the fate of **birds** and **bats** that fly into the blades of turbines. In northern Tasmania a number of **wedge-tailed eagles** have been killed as they glide on the wind currents (thermals) created by the wind turbines when looking for their prey.

Wind turbines are built in factories that use a lot of energy, creating **pollution**. However, this occurs only for a short time, unlike the pollution continually created when electricity is generated by coal.



1 Write your own opinion about wind turbines in the empty speech bubble.

Objectives

- To learn about solar energy
- To recognise the advantages and disadvantages of solar energy
- To understand the need to have a suitable home design to make the best use of solar energy

Prior knowledge

Students should be aware of solar energy through observation of panels on roofs, solar garden lights and roadside signs. They should understand that they experience the energy of the sun when they are outdoors, especially in summer.

Background

Solar power has become a much talked about and promoted form of energy. The first major use of solar cells was as a backup power supply to the Vanguard 1 satellite in 1958. Solar power is still used for telecommunication satellites and remote stations.

Today, Solar power is used to power offshore oil rigs, navigational buoys and railway crossings. It is also used for to provide electricity, heating and decorative light in domestic homes.

In the past, governments have promoted the use of solar energy through rebates and an ability to return excess power to the grid. However, it takes a number of years to make up the expense of installing solar panels.

Like other forms of renewable energy, interest and investment in solar energy has been increasing globally. Large solar plants are being constructed in the United States, Canada, Spain, Mongolia and China.

Starter activity

Lead a class discussion about what solar power is and how it is used. Some students might have solar power at home – ask them to share their knowledge about it. Encourage students to identify advantages and disadvantages associated with solar power. Record their ideas in a chart or mind map, and put it aside to revisit at the end of the unit.

Resource and Activity sheets

Read the Resource sheet 'What is solar power?', with the students. This sheet explores key features of solar power, such as where to place solar panels and the need to insulate and use energy-efficient appliances.

Provide students with the Activity sheet, 'Solar power'. They will need to refer to the Resource sheet, 'What is solar power?', to complete the questions. This activity will reinforce their understanding of the uses of solar power both domestically and in larger-scale applications. Discuss the use of solar power compared with diesel-powered generators or coal-fired power stations in terms of emissions of greenhouse gases.

Read the Resource sheet, 'Advantages and disadvantages (1)', with the class. Encourage the students to think of any other advantages or disadvantages they could add to the lists.

Have students complete the Activity sheet, 'Advantages and disadvantages (2)'. They will need to refer to the Resource sheet, 'Advantages and disadvantages (1)'.

Extension activity

Revisit the chart or mind map you made with the students for the Starter activity. Work with the class to add to the chart, and correct the previous information as needed. Ask students to consider the future of solar power in Australia, and to give reasons for their predictions. Add this information to the chart and display it in the classroom.

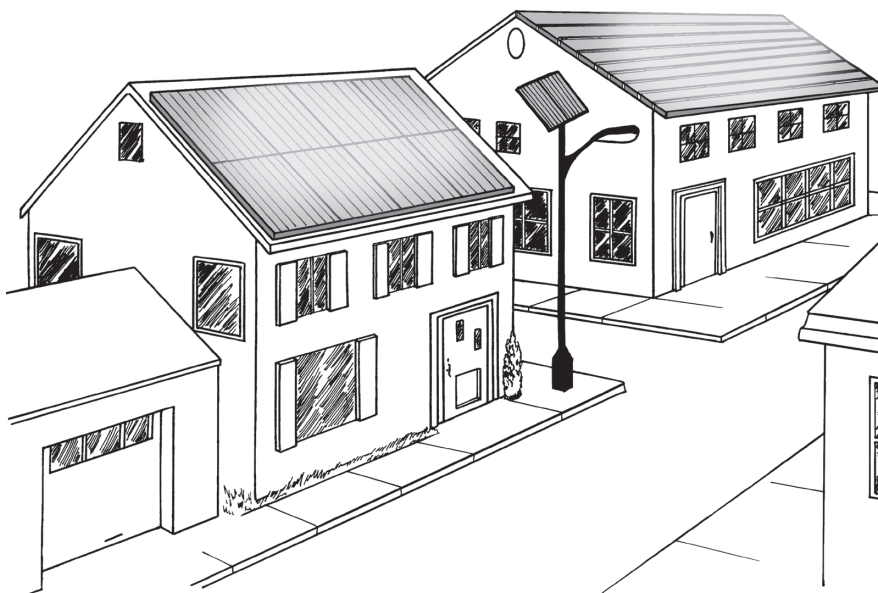
What is solar power?

Solar power is **renewable** energy from the sun which can be converted to electricity.

Most of Australia has a great deal of sunlight, but there is less in the southern and south-eastern areas. To make the most of solar power in homes, it is necessary to have energy-efficient heaters, insulation in the walls and ceilings, and solar hot water. **Solar panels** should be placed on north-facing roofs or in open spaces in order to generate the most power.

Case studies

- Solar power is extremely effective in small communities like **Windorah** in **Queensland**, which has a population of 100. Windorah has a group of five solar dishes which began generating electricity in 2008. Using them saves 100 000 litres of diesel each year, reducing greenhouse gas emissions by 300 tonnes per year.
- A Solar Concentrator Power Station is being built near **Mildura** in **Victoria** and is expected to be fully operational in 2013. The sunlight will be captured by curved reflective mirrors which track the sun throughout the day. The project will produce enough electricity to power more than 45 000 homes and reduce greenhouse gas emissions by about 400 000 tonnes per year.
- On average, **Mildura** receives 2938 hours of sunlight each year. There are 10.7 hours of sunlight per day in January and five hours per day in June.
- **Melbourne** receives 2200 hours of sunlight in an average year, with eight hours per day in January and four hours per day in June.



Solar power

Read the Resource sheet, 'What is solar power?', then work with a partner to answer the following questions.

 1 What is solar power?

2 Why is this method of power generation sustainable?

3 Which areas of Australia do not get a large amount of sunlight?

4 What else has to be done to make the most of solar power in homes?

5 Where should solar panels be placed? Why?

6 How many litres of diesel will be saved each year by using solar power at Windorah?

7 How many houses will the solar power station at Mildura supply?

8 What did you learn about the amount of sunlight received in Mildura compared with Melbourne?

9 How does the place you live compare with Mildura in terms of amount of sunlight received?

Advantages and disadvantages (1)

Advantages

- If you install solar panels, you can earn money by selling excess power to power companies.
- Solar power can be used for powering equipment like road signs, garden lights, football scoreboards and even the golf carts the Queen used in Canberra in 2011.
- Solar panels should last 25 to 50 years.
- Using solar energy creates jobs during design, manufacture, construction and operation.
- Solar power is a renewable energy source.

Disadvantages

- The sun does not shine all the time.
- It is expensive to buy and install solar panels.
- It costs extra to get the batteries you need to store power.
- Some people are concerned that hail and storms might damage the solar panels.
- Solar panels need to be cleaned.

Advantages and disadvantages (2)

Read the Resource sheet, 'Advantages and disadvantages (1)'.

-  1 Outline four advantages and four disadvantages of solar energy, in your own words. You could include labelled diagrams.

Advantages	Disadvantages

Objectives

- To understand what hunting and gathering means for a traditional society
- To recognise the sustainable use of the land by Indigenous people
- To become familiar with key terms such as: *Indigenous, inhabitants, remains, archeology, tradition, spirituality, prehistoric*

Prior knowledge

Students should be aware that there were Indigenous inhabitants in Australia before British settlement, and that there are still Indigenous people today. Some Indigenous people live traditional lifestyles today, or integrate traditional aspects into their lives.

Background

Early Indigenous Australians could only use the resources available in their local area. They spent a large portion of their daily lives acquiring enough food sources. Food also dictated the movement of the tribes.

Through their Dreamtime stories, Indigenous Australians understood sustainable use of the land, and practised this for thousands of years. Tribal cultural practices and their food sources differed according to their location. There was no cropping, although women in some areas replanted some roots. Kangaroos and other animals were not tamed or used as beasts of burden. Fish traps were built in coastal areas like Albany in Western Australian and along some inland rivers, but the catch was always sustainable.

Starter activity

Ask students about their understanding of the hunting and gathering lifestyle. Discuss how this compares with their own lifestyles. You could work with the class to research the food sources that were available to Indigenous Australians in your local area.

Resource and Activity sheets

Provide students with the Activity sheet, 'Indigenous Australians and sustainability'. Point out the timescale referred to – Mungo Man's age of about 60 000 years, compared to European occupancy of Australia a little more than 200 years ago. Discuss the differences between the methods of transferring knowledge and cultural practices compared with what happens today. Students will need a dictionary or Internet access to complete the definitions.

Read the Resource sheet, 'How did Indigenous people live? (1)' with the students. Discuss the sustainability of the Indigenous lifestyle. To extend the activity, you could make a class list of the food sources of Indigenous Australians, then compile a list of the food sources of the students.

Students will need a copy of the Resource sheet, 'How did Indigenous people live? (1)' to complete the Activity sheet, 'How did Indigenous people live? (2)'. Students will consolidate their understanding of the information on the Resource sheet by identifying statements that are true or false, and writing their own sentences about the lifestyles of the Indigenous Australians.

Extension activities

- Students could use the Internet to research Dreamtime stories, particularly those relating to finding food, and ensuring food resources were used sustainably.
- Have the students work in small groups to discuss how they could make their own food choices more sustainable. Each group could record their ideas as a mind map or create a poster.

Indigenous Australians and sustainability

Indigenous Australians are the original **inhabitants** of Australia and its nearby islands. The earliest human **remains**, dated at about 60 000 years old, are those of Mungo Man. He was found in Lake Mungo in New South Wales in 1974.

There is no written record of life in **prehistoric** Australia. Knowledge about this time is found in **archeological** evidence and in Indigenous oral **traditions**. Before European settlement, all Indigenous Australians based their lifestyles on their Dreaming beliefs. The Dreaming (also called the Dreamtime) stories connected humans, animals, fish, birds and the land. These connection brought a sense of caring about and **spirituality** towards the land and all the life it supported. The stories explained how to live sustainably so that the earth could support all kinds of life, both now and in the future.

 **1** What do the words in bold mean? Discuss them with a partner. Then write the words and their meanings below.

Word	Meaning
Indigenous	

How did Indigenous people live? (1)

- Indigenous Australians were hunter-gatherers who moved according to the availability of food. This way, their food supply was sustainable. The types of food available depended on whether they lived inland, near rivers or on the coast. The men hunted kangaroos, wallabies, echidnas and possums as well as snakes, lizards and birds such as ducks, especially along the Murray River. Spears and boomerangs made from local trees were used to catch and kill. Fish and eels were caught in traps in some areas, such as south-west Victoria.
- Fire was used to drive animals out and to 'clean up' the land. Fire was also used to keep travel corridors open and to encourage the growth of food plants such as bracken, daisy yams and grasses so that these resources could be sustained from year to year.
- Women and young children collected berries, yams, seeds, eggs and insects. If they lived near the coast, they collected oysters and mussels too. Seeds were ground between stones and cooked as cakes.
- Indigenous people knew how to hunt and gather sustainably so their food supplies would last. For example, they did not hunt young kangaroos or take more berries than they needed. However, it is thought that their practice of using fire did change parts of the landscape.
- Indigenous people understood the seasons and how they affected the food that was available. Along the Murray River they caught wildfowl and freshwater crayfish in spring, or netted fish in summer. They knew how to get groundwater in the desert from springs, rock holes and soaks, and dug wells to find water. Water sources were regarded as sacred. It was essential for people to remember where they were.



How did Indigenous people live? (2)

-  1 Read the Resource sheet, 'How did Indigenous people live? (1)'.
Decide whether the following statements are true or false.

Statement	True	False
Indigenous people were hunters and gatherers.		
The type of food available depended on whether they lived inland or on the coast.		
Men did not hunt.		
Fish were caught in traps.		
Indigenous people did not use fire.		
Women and children collected fruit, seeds and roots.		
Indigenous people knew how to look after their food supplies.		
Indigenous people tamed animals.		
Indigenous people did not know about the seasons.		
Water sources were sacred.		

- 2 Write four sentences to describe how Indigenous Australians lived in your own words.

What can we learn?

What can we learn today about living sustainably from traditional Indigenous practices?

-  1 Write what we can learn, and include specific examples. An example has been done for you.

Lesson	Example
Make sure there is enough food for other tribes, and for next year and the year after that.	Don't pick all the berries – leave some to go to seed so that new plants grow next season.

- 2 What are some of the advantages and disadvantages of living according to Indigenous practices, both in the past before European settlement and today?

Advantages	Disadvantages
In the past	In the past
Today	Today

Objectives

- To learn about the increasing demand for fish that results from an increasing human population
- To understand the difference between wild fishing and aquaculture
- To learn that conservation is essential for the sustainability of fishing
- To appreciate that while fishing is an important recreational activity for some people, other people rely on the food fishing provides in order to survive

Prior knowledge

Students should know that many people enjoy fishing for recreation and that fish is eaten regularly.

Background

As the population of the world increases so does the demand for fish, especially in developing countries.

Improved technology has made it easy to locate schools of fish, and led to improved fishing methods. Some of these methods, such as the use of cyanide, the blasting of reefs and bottom trawling cause great damage to the environment. Longline and purse-seine fishing result in large numbers of fish being caught, many as bycatch – fish that are not marketable. Many of the bycatch are dead when they are thrown back into the ocean. Overfishing has resulted in the significant reduction of many fish stocks, with some species such as southern bluefin tuna becoming critically endangered. Due to market demand, much illegal fishing occurs.

Taking too many fish commercially means there is not enough food for marine animals such as seabirds, turtles and sharks. Marine parks have been established to help protect fish species and to allow stocks to replenish, but fisheries say this affects their income. Governments have introduced quotas to try to make fish stocks sustainable.

Aquaculture has been used in many places around the world for prawns, oysters, mussels, barramundi and tuna to supplement the wild catch. However, aquaculture farms can cause other problems such as contamination from fish food and chemicals.

Starter activity

Discuss the different reasons people have for fishing. Bring out the idea that some people fish for fun, some for money and other to get the food they need to survive. Work with the students to define key fishing terms such as *wild catch*, *aquaculture*, *bycatch* and *overfishing*.

Resource and Activity sheets

Read the Resource sheet, 'Sustainable fishing', with the students. Review the meaning of each term shown in bold. Ask students to share any experiences they have with fishing and discuss the types of fish they eat.

Provide students with the Resource sheet, 'Bluefin tuna – wild catch or farm?' Students will need to refer to this Resource sheet to complete the questions on the Activity sheet, 'Fishing for bluefin tuna'. Extend the activity by leading a discussion about the negative impacts of aquaculture, such as the reduction of wild stock and the large quantities of anchovies and sardines that must be sourced to feed the farmed fish. Ocean and coastal pollution also occurs from chemical use and dissolved and solid fish waste.

Have the students read the Resource sheet, 'Orange roughy – overfishing and conservation'. They will need to refer to this Resource sheet to complete the questions on the Activity sheet, 'The orange roughy fish'. You could extend this activity by having students work in small groups to list arguments for and against the statement: *Fishing for orange roughy should be banned*.

Extension activity

- Students could use the Internet to find more species of fish and other marine species that are threatened, and possible solutions to these problems. The websites of organisations such as the Australian Marine Conservation Society, World Wildlife Fund and Greenpeace will be useful.
- Students could investigate the number of marine parks in Australia, map their locations and annotate the map to show which marine creatures live there.

Sustainable fishing

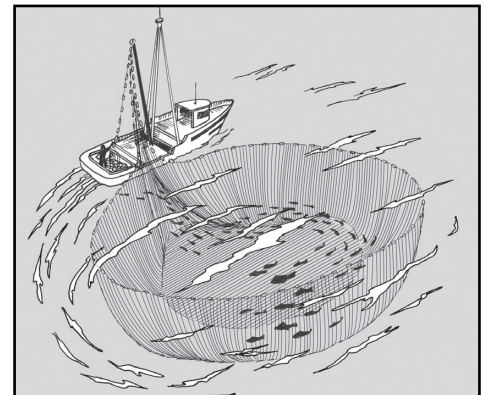
The health benefits of eating fish are well known and often promoted. People in many countries eat fish every day. Over 1 billion people throughout Asia depend on fish, including **subsistence** islanders in Papua New Guinea and people in the Philippines, who depend on fish for their animal protein. Many people are employed making and mending boats, nets and other equipment, as well as in fish processing.

In Australia, the fishing industry is number six on the list of the most valuable food-based **primary industries**. Almost 3.4 million Australians fish for **recreation** and contribute to the economy by buying fishing licences and fishing gear. Some Indigenous Australians fish for food, for **cultural practices** and to earn money.

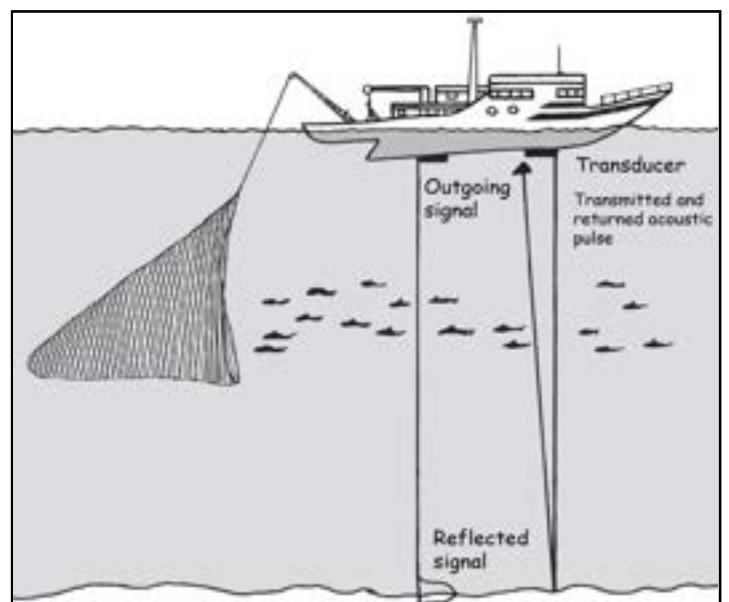
As the number of people in the world increases, so does the global demand for fish. This, combined with changes to methods of fishing, has resulted in **overfishing**. This means that more fish are being caught than the ocean can produce, so the overall number of fish is decreasing.

Bycatch is also a significant problem – this is the term for the fish and other marine creatures that the fisheries catch in their nets, but do not want and cannot sell. Often these creatures die.

Bluefin tuna is highly valued for its flavour, especially when served raw. It is overfished in the Pacific Ocean, and in most other parts of the world. One bluefin tuna caught on a longline off the Japanese island of Hokkaido was sold for \$396,000 in early 2011. That was a price of \$12,000 per kilo!



Purse-seining involves putting a long net around a group of fish, then closing the bottom of the net to capture them.



Midwater trawling involves dragging a huge net behind a boat. It catches a large number of fish. The use of technology such as electronic fish-finders makes it easy to work out where the fish are.

Bluefin tuna – wild catch or farm?

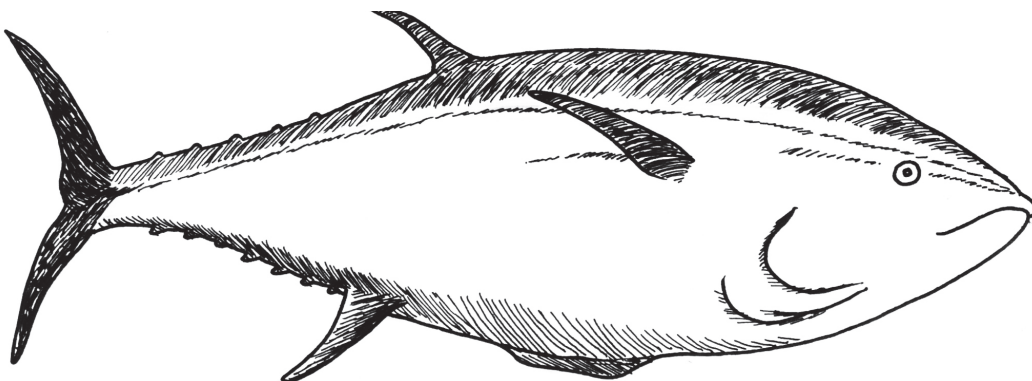
Tuna is an extremely popular fish, particularly **bluefin tuna**, and stocks in the Pacific and Indian Oceans are decreasing due to **overfishing**. With **wild catch** fishing methods, fisheries can catch entire schools of tuna.

Southern bluefin tuna are prized as an eating fish, especially for the traditional Japanese food **sashimi**. These tuna live for up to 40 years, grow to two metres long and weighing up to 200 kilograms. Due to the high demand for tuna, there are strict **quotas** in place to protect the population. Any fisheries catching more than their quotas are breaking the law. Illegal tuna fishing still occurs because people are prepared to pay a great deal of money for them.

Tuna farming at **Port Lincoln** in South Australia began in the early 1990s. By farming the tuna instead of catching them in the wild, the hope is that tuna populations can be sustained. However, farming can cause other problems such as contaminating the oceans with farm chemicals and the food fed to the fish. Controlling fish diseases is also more difficult.

The following information describes how **southern bluefin tuna** are farmed.

- The tuna are caught by purse seine net in the Great Australian Bight in January.
- They are transferred live to a towing cage.
- The cage is towed very slowly for about 20 days to near Port Lincoln.
- The tuna are swum into smaller cages and bred in captivity.
- They are fed six days per week, weather permitting.
- When there is a growth rate of 40%, usually in late winter, they are harvested.
- They are processed and placed in boxes, frozen or with chill packs.
- The fish are then flown to Japan, South-East Asia and Europe.

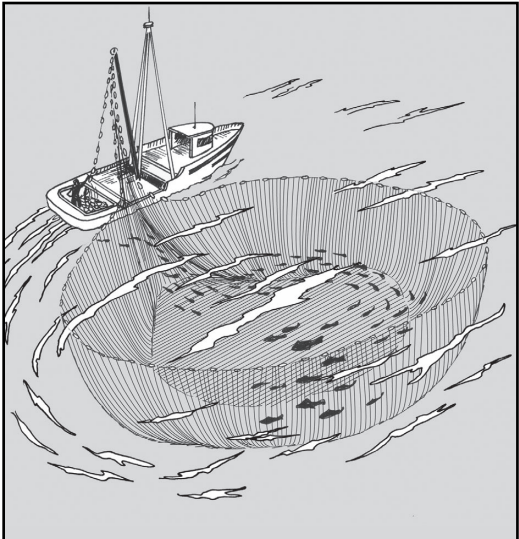


Fishing for bluefin tuna

 1 Use the Resource sheet ‘Bluefin tuna – wild catch or farm?’ to decide whether the statements in the table are true or false.

Southern bluefin (SB) tuna	True	False
SB tuna is in great demand.		
SB tuna is not overfished.		
Illegal fishing is not a problem.		
The Japanese market pays large amounts of money for SB tuna.		
SB tuna farming began in 2010.		
Farming will definitely solve the problem of sustaining SB tuna stocks.		
SB tuna are caught by purse seine nets.		
The fish are fed once a week.		
SB are harvested in January.		
The fish are chilled or frozen for export.		
The markets for SB tuna are Japan, Europe and South-East Asia.		

2 Does farming help with the sustainability of southern bluefin tuna? Explain your answer.



Using a purse seine net

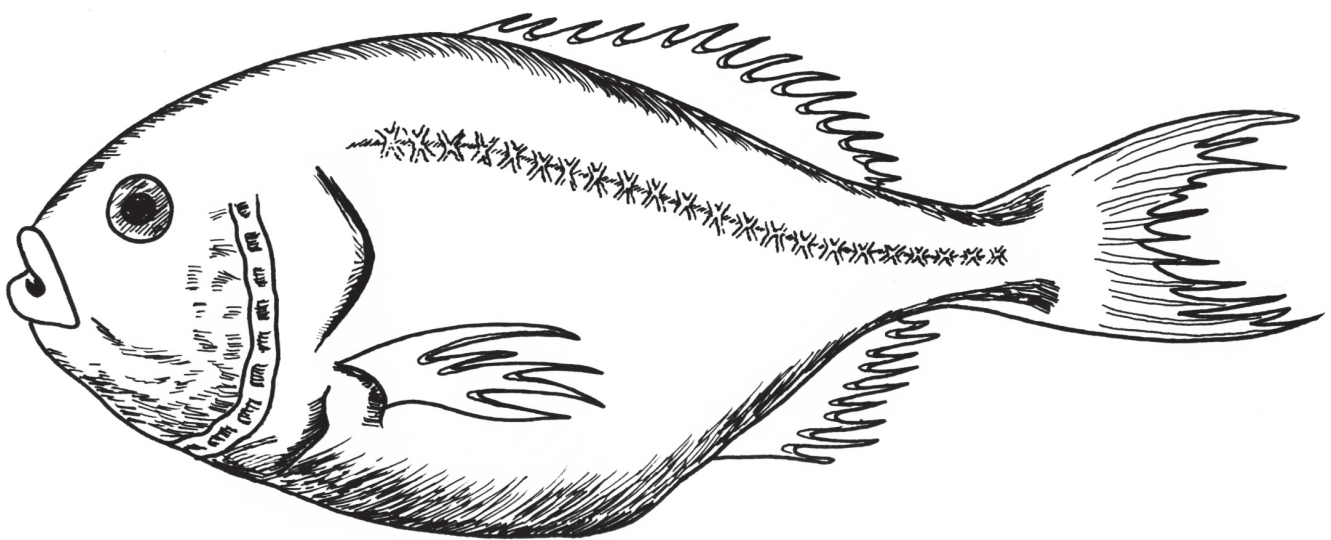
Conserving the orange roughy

Orange roughy is a bright red fish with a large, bony head and big eyes. It lives in cold, deep waters in the **Atlantic, Pacific** and **Indian Oceans**, especially around seamounts which are volcanic mountains under the sea.

This species may live beyond 150 years. It matures between 20 and 40 years of age and feeds on prawns, fish and squid. Very young fish are rarely caught by fisheries, but larger specimens can be caught well before they can reproduce at about 30 years of age.

The **New Zealand** orange roughy fishery is the oldest and largest in the world. It was established in the late 1970s and production peaked at approximately 22 000 tonnes in the mid-1980s. That figure has now been reduced to about 8650 tonnes.

In New Zealand, orange roughy fishing is managed under a **quota system** in a 200-nautical-mile exclusive economic zone, in order to help keep stocks at a sustainable level. Measures are being taken to restrict where fish can be caught, and **marine national parks** established to provide a safe place for fish to grow and breed. Similar restrictions apply to orange roughy fishing in southern Australian waters.



The orange roughy fish

Read the Resource sheet, ‘Conserving the orange roughy’. Then work with a partner to answer these questions.

The orange roughy fish

 1 What does it look like?

2 Where does it live? _____

Its life

3 How long does it live? _____

4 What does it eat? _____

5 What does it mean for the future if orange roughy are caught before they are 30 years old?

Fishing

6 Which country has the oldest orange roughy fishery? _____

7 When did fishing for orange roughy at this fishery begin? _____

8 What has happened to the number of orange roughy being caught?

Sustaining the numbers of orange roughy

9 What does ‘quota’ mean in terms of commercial fishing?

10 What have New Zealand and Australia done to try keep up the stock of orange roughy?

Objectives

- To recognise that water is essential for survival
- To understand how people, plants and animals rely on water
- To learn about the need to conserve water to ensure sustainability of food supplies and communities

Prior knowledge

Students have knowledge of some of the uses of water and have some understanding of droughts and floods.

Background

Water is essential for life. All the ecosystems on land and within oceans and seas rely on it. Access to water is also critical for the production of food and industrial products. This is why some rivers have been dammed to provide a continual supply. Due to people's need for water, the welfare of the environment and habitats for wildlife have at times been overlooked.

In recent years, long-term droughts have been followed by major floods, such as those in Queensland and Victoria in 2011. These floods replenish rivers, dams and groundwater as well as adding fertile silt to flood plains. However, the human cost is enormous through loss of life, livelihood, homes and possessions, as well as the effect on the Australian economy.

The Murray-Darling Basin is important to the 3 million people who live in and around the basin and depend on the water it provides. The Murray-Darling Basin is also extremely important to the rest of Australia, as one-third of the food supply is produced in the region. This contributes \$9 billion to Australia's economy through food supply and exports such as cotton and rice.

Starter activity

Brainstorm with the class a list of issues related to water; for example, effects of droughts and floods, pollution of waterways, global problems such as lack of drinking water, local water restrictions, water recycling. Guide students to realise that water management is an extremely important issue, both now and in terms of sustainability for the future.

Resource and Activity sheets

Read the Activity sheet, 'Fact file – water', with the students. Students will unjumble words to complete the water facts. Ask the students to identify any facts they found surprising, and discuss why.

Provide the Resource sheet, 'Case study – Murray-Darling Basin'. Discuss what the map shows. You could have students identify the region on a complete map of Australia to assist their understanding of its size. Review the key features of maps with the class, and explain the meaning of *tributary*, *source* and *mouth* in the context of a river system.

Have the students complete the Activity sheet, 'Using the Murray-Darling Basin map'. They will need to refer to the Resource sheet, 'Case study – Murray-Darling Basin'.

Provide the Resource sheet, 'What can be done to make the Murray-Darling Basin sustainable?' and read it with the students. Discuss the role of government in water management, and the importance of community concerns. Introduce the concept of irrigation and give examples like Mildura which can exist in an extremely dry area and have a thriving agricultural industry because of water from the Murray River.

Ask students to complete the Activity sheet, 'Water supply issues'. You could have them use the Internet to conduct further research into the issues raised. Encourage them to think about these issues and what they can do to help make water sustainable.

Extension activity

- Collate the water questions students composed on the Activity sheet, 'Water supply issues'. Have them work in small groups to answer the questions.
- Students could use the Internet to investigate the use of water for growing crops like cotton and rice. Discuss whether this use of water can be justified. Students should identify which crops in Australia are sustainable and give reasons. Encourage them to consider the issue of cost in terms of equipment, infrastructure and time involved in changing agricultural products.

Fact file – water

-  1 Unjumble the words in brackets and write them correctly to complete these water facts.

Fresh _____ (tarwe) is essential for sustaining all forms of _____ (eifl). Humans, both as individuals and communities, depend on it. So do the _____ (snmiala) and _____ (slnpat) of the earth's ecosystems.

Climate _____ (heancg) affects where rain falls and how much _____ (nari) falls. It also leads to more storms and droughts. As the population increases, the demand for water will _____ (ncesreia). Humans need 2.5 litres of water per day just to live. Food production and industry requires a great deal more. In some countries, like Pakistan, there are people who must _____ (wkla) long distances each day in order to find clean _____ (iidxgnnr) water. Other people do not have access to clean water at all.

Australia is the driest inhabited continent in the _____ (dwrlo). Fresh water is, on average, very scarce. This means that _____ (eeorruccs) must be looked after very carefully. South Australia is the _____ (trides) state in the Australia.

Large _____ (osnmuat) of water are used in processing _____ (oofd) and in manufacturing. For example, it takes three litres of water to produce one _____ (terli) of bottled water.

Water must be managed carefully if we want to maintain our _____ (uplsyp) of crops, fruit and vegetables. Water should be used sustainably to ensure the health of _____ (storesf), wetlands, trees, fish and _____ (ridsb), as well as provide enough water for the _____ (uteruf) needs of urban communities.

Case study – Murray-Darling Basin

The Murray-Darling Basin covers one-seventh of Australia. It produces one-third of Australia's food supply. Australia's three longest rivers – the Murray River, the Darling River and the Murrumbidgee River – are the focus of the basin. Their **tributaries** are extremely important for supplying water to communities for everyday use and agriculture.

MURRAY-DARLING BASIN



Using the Murray-Darling Basin map

Use the map on the Resource sheet, 'Case study – Murray-Darling Basin', to answer the questions.

-  1 Which states contain the Murray-Darling Basin?

- 2 Where is the source of the Murray River? _____
- 3 Which states have the Murray River as their border?

- 4 Where do the Murray River and Darling River meet? _____
- 5 Which town is closest to the mouth of the Murray River? _____
- 6 Use the scale on the map to work out how long:
a) the Darling River is between Bourke and Wentworth. _____
b) the Murray River is between Echuca and Mildura. _____
- 7 How many tributaries of the Murray River are shown on the map? _____
- 8 Which state does not have a tributary of the Murray River? _____
- 9 Highlight the Hume Dam and Dartmouth Dam on the map. Explain why these dams have been built.

- 10 Use the scale on the map to write two more distance questions about the map.
Then challenge a partner to work out the answers. One example has been done for you.
a) How long is the Murrumbidgee River between Wagga Wagga and where it meets the Lachlan River? _____
b) _____

c) _____

What can be done to make the Murray-Darling Basin sustainable?

The federal and state governments are trying to ensure that water is used sustainably in the Murray-Darling Basin. In 2011, the second draft plan for the basin was released, identifying ways to reduce water use. People who depend on the water for their communities and for agricultural production are concerned that too much water will be used for wetlands, trees and birds.

Here are some concerns about the water supply.



Should we grow cotton, which requires huge amounts of water?



If the river is low, how can I operate tourist boats?



Will people come to camping grounds, caravan parks and motels if we can't water the gardens?



Should I pull out the grape vines and citrus trees that I irrigate?



What will happen to the river, swamps and wildlife if there is not enough water?



I live in Adelaide. The Murray-Darling Basin is our only source of drinking water.

Water supply issues

Work in pairs to discuss these questions, then write your answers below.

-  1 How can we save water at home and in the community?
- ---
- 2 Should we build more dams to catch water when rainfall is high? Why?
- ---
- 3 Do water restrictions make people think carefully about water use? Explain your answer.
- ---
- 4 How can water recycling help with sustainability? How can recycled water be used?
- ---
- 5 Do you think it is important for rivers and wetlands to have a constant supply of water so they can be sustainable into the future? Give two reasons for your answer.
- ---
- 6 It is difficult for farmers to change the types of crops they grow. Suggest why.
- ---
- 7 Write two more water questions you would like to have answered.
- a)

- b)

Objectives

- To recognise the significance of coral reefs
- To learn about the locations and types of coral reefs
- To understand the use and abuse of coral reefs
- To appreciate the need for coral reefs to be sustainable

Prior knowledge

Students should be aware that the oceans contain a huge variety of life, which is important for human consumption, recreation and tourism, and essential to preserving ecosystems.

Background

Coral reefs are very complex and fragile ecosystems that exist in the oceans. Their ecological significance is often compared with that of tropical rainforests on the land because of the diversity of species found within them. Coral reefs require warm, clear water less than 60 metres deep. Their future is threatened by fishing, tourism, oil spills, pollution, agricultural chemical run-off, silt and sewage. Climate change also poses a threat to the sustainability of coral reefs.

Starter activity

Ask students to list the kinds of creatures they associate with the ocean, and specifically with coral reefs. Encourage students to share any experiences they have with coral reefs. You could use a mapping application such as Google Earth to view coral reefs.

Resource and Activity sheets

The Activity sheet 'What are coral reefs?' introduces students to the structure of reefs and the kinds of animals that live there. Discuss the meaning of the terms in bold. Students could use the Internet to help them complete the last activity. To extend the activity, you could use the key terms on the sheet to start a class pictorial coral reef glossary.

On the Activity sheet, 'Types of reefs', students complete cloze exercises to explore the three main types of reefs, and the underlying structure and examples of each type.

The Activity sheet, 'Where are coral reefs?' provides an opportunity for students to practise their mapping skills. Students will need an atlas or Internet access, and a blank world map, the completed Activity sheet, 'Types of reefs', to complete this activity. A map is provided on the CD-ROM in the Artwork > Unit 9 folder.

The Activity sheet, 'Life on a coral reef', provides an opportunity for students to investigate the different kinds of marine creatures that live around coral reefs. Students will need to use the Internet to label marine creatures shown in the illustration, then draw and label a further list of marine creatures. You could challenge students to find the most unusual coral reef creature they can to add to the illustration.

On the Activity sheet, 'Protecting reefs', students explore threats to the reef, and efforts to protect and restore reefs. Explain that carbon dioxide from power stations and motor vehicles causes warming of the atmosphere, which in turn threatens coral reef habitats which take thousands of years to form.

Emphasise sustainability when students complete the sentences on the Activity sheet, 'What can be done to ensure that coral reefs are sustainable?' Discuss the idea that all parts of the environment are connected, and that people can't protect an area like a coral reef without considering what happens to the water they use and the rubbish they generate in their own homes.

The Resource sheet, 'How are coral reefs used?', informs students about the ways coral reefs are used today, and the increasing number of visitors to coral reefs. Encourage them to think of other ways that people use coral reefs.

Have students complete the Activity sheet, 'How can people help to save the reefs?' They may find it useful to refer to the Resource sheet, 'How are coral reefs used?' Encourage them to use the space at the bottom of the sheet to list other ways different people can help to save coral reefs.

Extension activity

Have students work in small groups to create a pictorial glossary of marine life around coral reefs. They could focus on the creatures in a specific marine park, such as the Great Barrier Reef Marine Park or Ningaloo Marine Park. To extend the activity, have them identify food chains or webs within the marine parks.

What are coral reefs?

 1 Use the word bank to complete the information below.

Coral reefs are often called ‘rainforests of the _____’ because they form some of the most _____ **ecosystems** on earth.

Coral reefs are home to 25% of all **marine** _____.

Tropical rainforests contain half of all the species and plants on earth.

Coral reefs are formed from coral polyps – tiny _____ that live in colonies. When the polyps die, they leave behind a hard structure made of **limestone**. New **hard corals** such as _____ and elk corals grow on the old corals, building up the reef.

Soft corals do not build reefs but _____ themselves to the hard corals. Corals are found in oceans that do not contain many **nutrients**.

A large variety of organisms _____ in and around the reefs, including fish, seabirds, sponges, corals, _____, worms, shrimp, spiny lobsters, crabs, clams, starfish, sea urchins, sea cucumbers, sea squirts, sea turtles and sea _____.

 2 Explain why coral reefs are called ‘rainforests of the sea’.

Word bank

- animals
- attach
- brain
- diverse
- jellyfish
- live
- sea
- snakes
- species

3 What percentage of all marine species live in coral reefs? _____

4 How could agricultural nutrients being washed into the sea affect coral reefs?



Types of reefs

 1 Use the word banks to complete the information below.

There are three main types of reefs: **fringing reefs**, **barrier reefs** and **atolls**.

Word bank

shallow common Ningaloo largest islands

A **fringing reef** grows next to the shore, or borders it with a _____ lagoon in between. It is the most _____ kind of reef. _____ Reef, which lies off Western Australia, is Australia's _____ fringing reef. Fringing reefs are found near many of the small _____ of Papua New Guinea, in the Maldives, throughout the Pacific Ocean and in Florida Keys.

Word bank

separated outer largest 2900 Red coastline Bahamas

A **barrier reef** is _____ from mainland or an island shore by a deep lagoon. These reefs grow along the _____ edges of continental shelves. The Great Barrier Reef is the largest example, and also the _____ coral reef system in the world. It has more than _____ individual reefs stretching for over 2600 kilometres off the coast of Queensland. The New Caledonian Barrier Reef stretches for 1500 kilometres. In the _____ Sea, there are fringing reefs 6000 years old, located around a 2000 km _____. The Bahamas Barrier Reef is located between Andros and Nassau in the _____.

Word bank

central enclose Indian level volcano

Atolls, such as the Bikini Atoll, are almost circular reefs that _____ lagoons. Atolls begin growing around the edge of a _____. The atoll grows upward as the volcano sinks or the sea _____ rises. Eventually the volcano is under the ocean's surface, leaving the reef enclosing a _____ lagoon. Atolls can be found in the Maldives in the _____ Ocean as well as in the island groups of Kiribati, Tuvalu and Tokelau in the Pacific Ocean.

Where are coral reefs?

You need a world map, an atlas or Internet map search, and the completed Activity sheet, 'Types of reefs'.

Coral reefs need special conditions to exist. Coral lives in both **temperate** and **tropical** waters. Shallow reefs form between the **latitudes** 30°N and 30°S. The maximum depth for tropical corals to grow is about 60 metres, because the special **algae** corals contain need sunlight. The algae use sunlight and the polyps' waste products to make oxygen and food for the corals to grow. The water must be clean and clear so light can reach the coral.

The best seawater temperatures for growth are 24–31°C. This is why there are no reefs near the mouths of major rivers like the River Ganges and the Indus River, which carry very dirty water. Coral reefs don't grow in places where rivers meet the ocean. This is because the rivers put nutrients, pollution and fresh, colder water into the ocean.

 **1** On your world map, mark and name the countries and reefs mentioned on the Activity sheet, 'Types of reefs'.

2 Now add these reefs to your map:

- | | |
|---|---|
| <ul style="list-style-type: none"> • Mesoamerican Barrier Reef System, stretching from Isla Contoy at the tip of the Yucatán Peninsula to the Bay Islands of Honduras • Aldabra Atoll, Seychelles | <ul style="list-style-type: none"> • Andaman Sea Coral Reefs, off the coast of Myanmar • Raja Ampat Islands, West Papua • Matangi Island, Fiji |
|---|---|

3 Describe where the coral reefs are found in terms of lines of latitude, and the countries and continents they are close to.

Latitude _____

Countries _____

Continents _____

Life on a coral reef

 **1** Label these creatures on the illustration. Use the Internet to help you identify them, and colour them appropriately.

- shark
- anemones
- starfish
- nudibranch
- lionfish
- moral eel
- brain coral
- sea horse

2 Now use the Internet to find out what these creatures look like. Draw them on the illustration, and colour them appropriately.

- clown fish
- sea turtle
- giant clam
- sea urchin
- blue dory
- sea squirt
- spiny lobster
- christmas tree worm



Protecting reefs

 1 Unjumble and write correctly the words in brackets to complete the information below. The first letter is underlined.

Coral reefs are very **complex** and **fragile** _____ (scosteemsy). As they are **temperature sensitive**, they will be affected by (sgriin) sea temperatures caused by _____ (mitcael) change – the _____ (loacr) can die and become white, or bleached.

Other threats come from:

- **overfishing**, especially **blast fishing** in Indonesia and the Philippines
- **fishing** by trawling, which can _____ (kreba) coral and also _____ (stacche) fish that cannot be sold
- **cyanide fishing**, which stuns fish so they can be collected and sold as _____ (uuarimq) fish
- **run-off** containing agricultural _____ (leihcamsc), silt, and other pollution
- people taking _____ (lesshl) and coral for souvenirs
- careless **tourists** causing damage by _____ (agklinw) on the reef
- the _____ (abtos) of tourists and sightseers running onto the reef, or _____ (mgaaindg) coral with anchors.

Fortunately, more _____ (eiarnm) protected areas are being established around the world. These areas are like national _____ (rpask) on the land – they encourage responsible _____ (shinigf) and protect the _____ (tthaiasb) by restricting and monitoring activities that can cause damage.

Scientists are also experimenting with restoring coral _____ (esfer) through coral aquaculture, which is also known as coral gardening. They grow coral seeds and when they are large enough, _____ (traplen) them on damaged reefs. Local _____ (elepop) are paid to look after the coral as it grows. Another technique used is transplanting coral from _____ (talhhye) reefs to damaged reefs.

What can be done to ensure that coral reefs are sustainable?



1 Add the missing vowels to complete these tips about looking after coral reefs.

It is important to cr__ __ te marine n__ tional p__ rks like those around Apo Reef in the Philippines and the Great B__ rr__ __ r Reef in Australia

More marine sp__ c__ es should be protected by international l__ w.

Don't eat sh__ rk f__ n soup, t__ rtle eggs or turtl__ meat.

Check where __ q__ __ rium fish come from and don't buy f__ sh caught in c__ r__ l reefs.

As a t__ __ rist, avoid b__ y__ ng souvenirs made with sh__ lls, dried s__ __ h__ rses or d__ __ d corals.

Tell people n__ t to w__ lk on corals as they are l__ v__ ng cr__ __ t__ res and can break and d__ __ .

R__ duc__ your consumption – think 'Reduce, reuse and r__ cycl__'.

B__ y l__ ss plastic and get rid of it responsibly, so it doesn't end up in the oc__ __ n.

K__ __ p the environment cl__ __ n – don't leave rubbish, and clean up any r__ bb__ sh you find on the b__ __ ch.

B__ responsible in the way you use w__ t__ r and with the ch__ m__ c__ ls you use. These chemicals can get into the w__ t__ rw__ ys and oceans.

Only __ __ t fish and s__ __ f__ __ d that comes from s__ st__ __ nable f__ sh__ ries.

How are coral reefs used?

People are fascinated by coral reefs and the varied and colourful marine life that lives there. Many people have made a living from coral reefs through **fishing** and, more recently, **tourism**. However, these activities, along with natural causes like **floods**, storms and **temperature changes**, have created problems. More and more people are visiting the reefs for a variety of reasons.



As a tour operator I take people to see the reef.



I am a marine biologist and study the animals and plants of the reef.



My husband and I want to spend a holiday exploring the reefs.



I look after people when they go diving and snorkelling on the reef.



My occupation is fishing. I need to catch a lot of fish to sell.



I like to go to the reef to fish for recreation.



My job is to look after the reef, as part of it is a marine park.



I sell colourful fish for fish tanks.

How can people help to save the reefs?

 1 Read the Resource sheet, 'How are coral reefs used?' Then use your knowledge about coral reefs to describe how these people can help to save the reefs.

A person on a tour holiday at Apo Reef in the Philippines _____

A fisherman _____

A teenager concerned about the environment _____

A person running a restaurant _____

A pet store owner _____

An international lawyer working for the government _____

A marine biologist _____

A dive boat operator _____

A shopkeeper selling souvenirs _____

Objectives

- To understand the importance of forests for the production of oxygen
- To appreciate the significance of forests providing a sustainable habitat for animals, food and medicinal plants
- To learn that demand for timber and other products creates problems for the environment
- To learn that the forest products people choose to buy can contribute to the conservation and sustainability of forests

Prior knowledge

Students should be aware that people use forest products every day.

Background

Forests in all regions of the earth are extremely important as trees take in carbon dioxide and produce oxygen. With the climate changing and the need to reduce carbon emissions, the conservation of forests is crucial to sustainability. The health of the planet is not the only factor – we also need to manage the ever-growing demand placed on forest products by our increasing population. This increasing demand has led to the need for plantation timber and the preservation of old growth timber. Rainforests have been particularly vulnerable to wealthy nations using the land for oil palms and rubber plantations. Illegal logging relies on corruption in countries like Indonesia and is aided by the demands for forest products from other countries. Environmental organisations like the World Wildlife Fund (WWF) are trying to promote sustainable forestry in many countries.

Starter activity

Work as a class to make a list of the ways people use forests, and the ways people use products made from forest resources. Then discuss the sustainability of the products made from forest resources, and any alternative products people can use that have a lower environmental impact.

Resource and Activity sheets

Read the Resource sheet, 'Forestry (1)' with the students. Discuss the role of forests in sustaining biodiversity and providing employment for rangers and workers in the timber and tourism industries. Ensure students understand the words in bold, especially the difference between native and plantation forests.

Have students complete the Activity sheet, 'Forestry (2)', which provides an opportunity for students to review and summarise the information from the Resource sheet, 'Forestry (1)'. Students will need this Resource sheet to complete the activity.

On the Activity sheet, 'Protecting East Gippsland's old forests', students investigate an Australian example that highlights the importance of conserving and using old-growth forests. The students will use a word bank to complete the information.

Provide students with the Resource sheet, 'Rainforests in Kalimantan, Indonesia', which shows the differences between original rainforest and logged rainforest. Encourage students to suggest reasons why people log rainforests.

Students could work in pairs or small groups to discuss and complete the Activity sheet, 'Logging the forests'. Students will find it useful to refer to the Resource sheet, 'Rainforests in Kalimantan, Indonesia'.

Extension activity

- As a class, investigate how people can find out whether the forest products they use come from sustainable sources.
- Discuss the environmental problems and social issues associated with illegal logging and what can be done to prevent it.



Forestry (1)

Forests are extremely important to the health of the earth. They absorb **carbon**, which is emitted when **fossil fuels** are burned. Forests produce oxygen, which people and animals need to live. They also provide wood, which people use to construct houses and other buildings, and to manufacture products such as furniture and paper.

Forests provide a habitat for animals, birds and a great variety of plants that provide fruit and medicines.



About 19% of Australia is covered with forest. **Native forests** cover approximately 147 million hectares, and **plantation forests** cover approximately 2 million hectares. Plantation forests contain trees such as pines and blue gums, for commercial use.

Forests must have water, soil and nutrients to survive. Many forest **catchments** provide an important source of drinking water for towns and cities.



Some people in Tasmania and Victoria do not want **old-growth forests** logged because **biodiversity** is lost, as are the habitats of birds, animals and plants.

Many old-growth forests are in national parks and state parks. Picnic areas, camping grounds, bushwalking tracks and nature walks are available in many parks.

Forestry (2)

 **1** Use the Resource sheet, 'Forestry (1)', to complete the boxes below.

Forests absorb...

Forest produce...

Wood is needed...

Forest habitats...

Plantation forests are...

Forest catchments...

Biodiversity is lost by...

People can use national parks and state parks for...

2 Why is it important to look after forests and make sure they are sustainable and renewable resources?

Protecting East Gippsland’s old-growth forests

-  1 Use the word bank to complete the information about the importance of old-growth forests in East Gippsland in Victoria.

Word bank

timber plants employment animals untouched recreation
conservation timber protection temperate visit

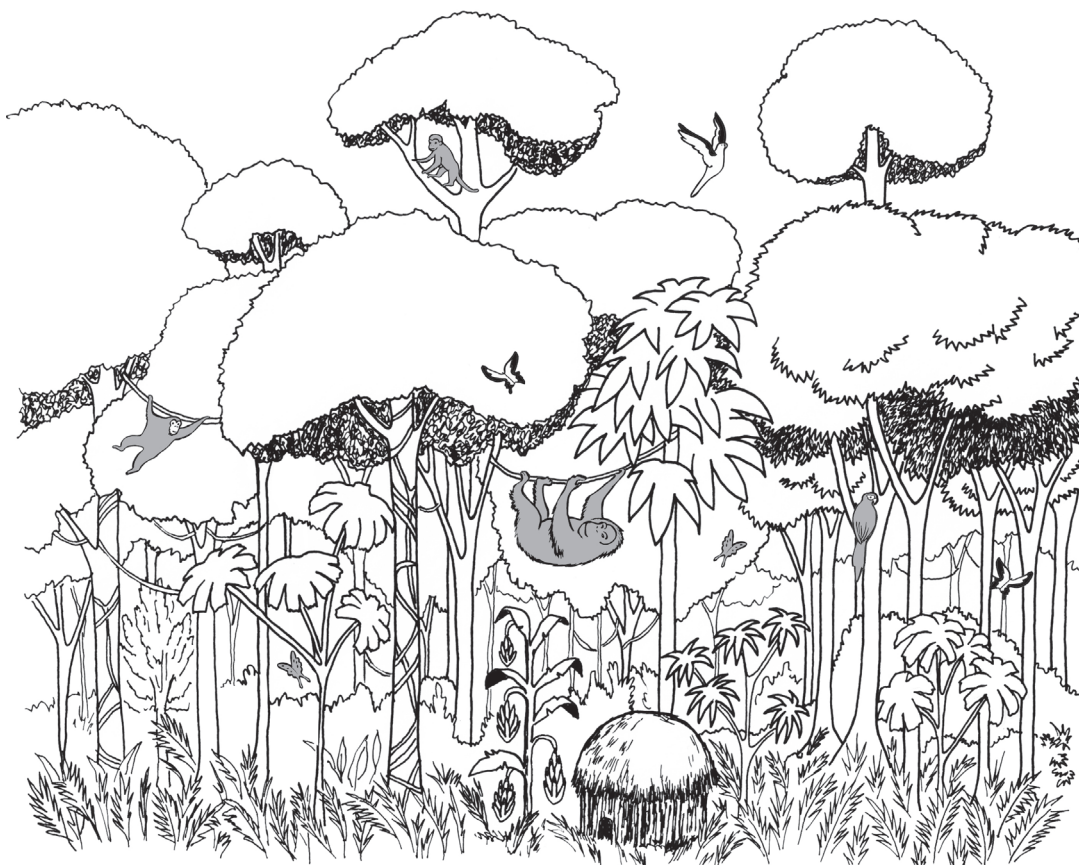
East Gippsland has warm and cool _____ rainforests, which are home to some unique _____ and _____. The old-growth forests are magnificent to look at, and many people want them to remain _____.

People, especially those from urban areas, like to _____ these forests for _____. Much of the area is protected by national parks and _____ reserves, with large new areas being added by the Victorian government. The new areas will help with the _____ of animals including the Long-footed Potoroo, Spotted-tailed Quoll, Powerful Owl and Large Brown Tree Frog. Native animals and birds will be able to move around and breed.

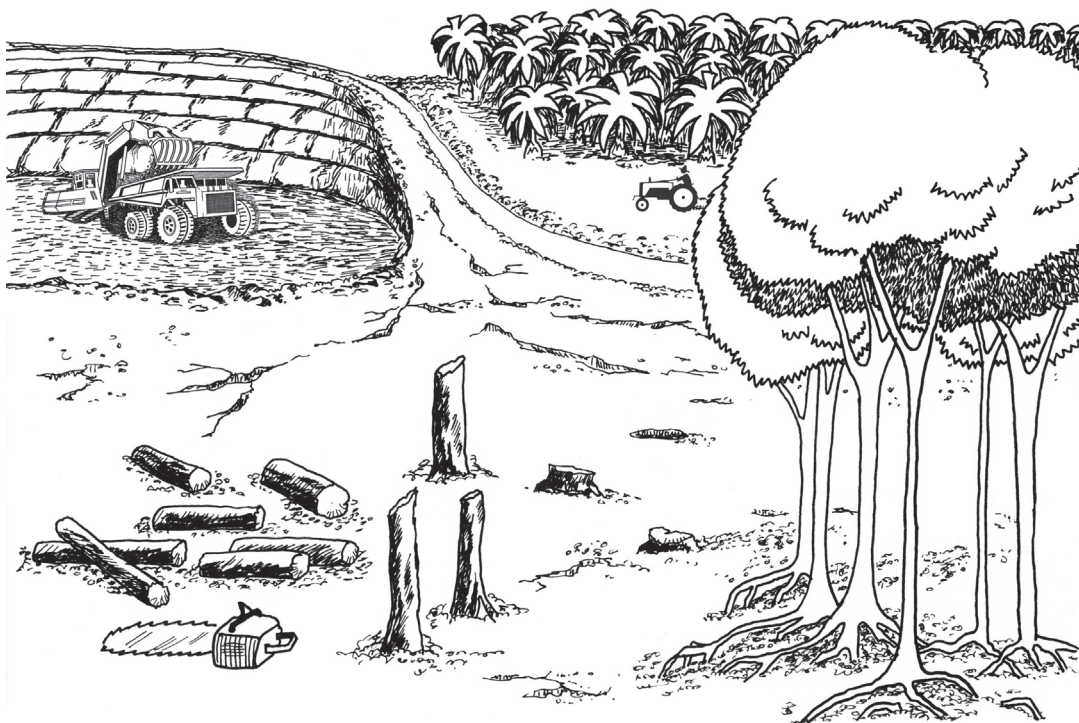
Some areas will still be used for _____, which is important for _____ and providing resources to make products that people need.



Rainforests in Kalimantan, Indonesia



Rainforest with many resources and habitat for wildlife



After logging

Logging the forests

Indonesia has the third largest tropical rainforest in the world.

Examine the Resource sheet, ‘Rainforests in Kalimantan, Indonesia’.

-  1 List the differences between the original rainforest and the area after logging.

Original rainforest	After logging

- 2 List the ways these areas can be used by people.

Original rainforest	After logging

- 3 Describe the impact that logging has on people, animals and the environment, both locally and globally.

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Achieve! Geography Sustainability in the Asia Pacific

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This title will assist teachers to develop lessons regarding the cross-curriculum priorities – *Sustainability*, and *Asia and Australia's engagement with Asia* – from the Australian Curriculum. Students develop their geographic understanding and inquiry skills as they learn about alternative sources of energy production including solar, hot rock, and wave power. The work sheets also focus on how communities in the Australasian region respond to these new production forms as well as the impact they have on the environment and peoples' everyday lives.

The CD-ROM contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other work sheets. These documents can be easily edited to suit the needs of your students.

Topics include

Sustainable energy
Wave power
Thermal energy
Wind power
Solar energy

Indigenous use of the land
Sustainable fishing
Water
Coral reefs
Forests

Associated titles in the *Achieve!* Instant Lessons series

- *Achieve!* Geography series
 - Indigenous contexts
 - Physical processes
 - Global issues
 - Sustainable development
 - Rivers: From source to sea

Instant Lessons

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